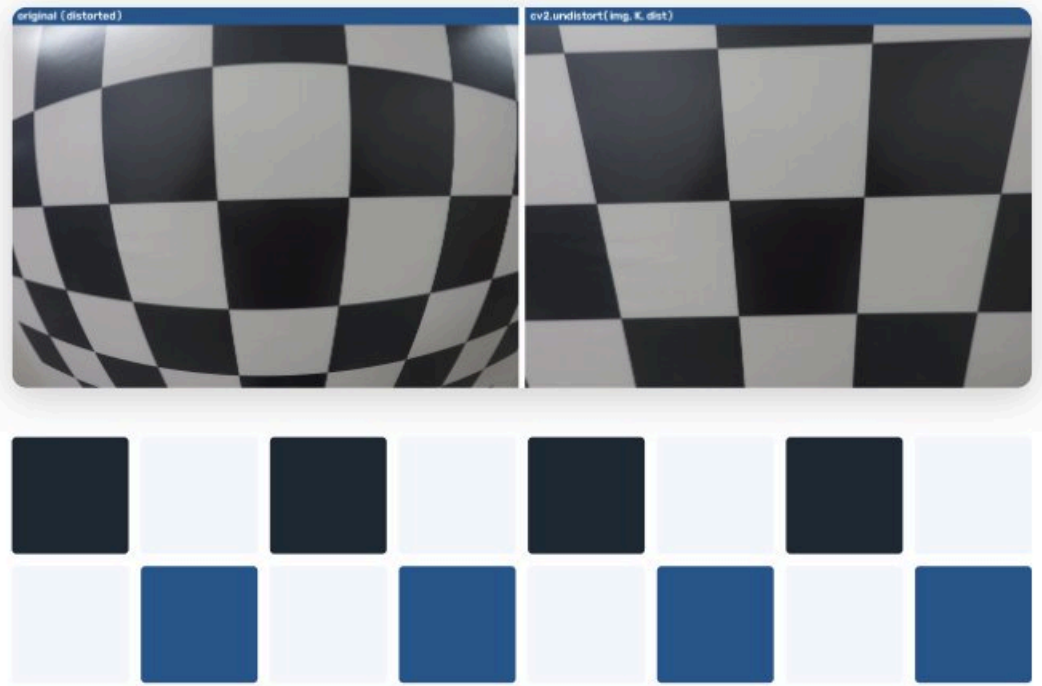
LECTURE 6

Camera Calibration

From pixels to the real world: camera models, lens distortion, calibration & perspective transforms

Pinhole, wide-angle, fisheye, stereo & Raspberry Pi cameras · hands-on with OpenCV · Lab 6



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WHERE WE ARE

From lane pixels to real-world measurements

Lecture 5 · you can already

- Select lane paint by **colour** and **region**
- Find edges with **Canny**, lines with **Hough**
- Draw two straight lane lines on **video**

…but everything is in **pixels**, and straight lines only.

Lecture 6 · today

- How a camera turns **3-D points into pixels** (pinhole model, **K**, [**R** | **t**])
- Why lenses **distort** and the 5 numbers that fix it
- **Calibrate** any camera with a chessboard (Zhang's method)
- Fisheye, ChArUco, stereo & Raspberry Pi cameras
- **Perspective transform** → bird's-eye view → lane **curvature** in metres







To steer, a car must measure how much its lane **curves**, which means looking at the road from above. That warp is only correct after we remove **lens distortion** (Udacity).

Learning outcomes

By the end of this lecture and Lab 6 you should be able to:

1. Write the **pinhole projection** $s \cdot x = K\,[R \mid t]\,X$ and explain every intrinsic and extrinsic parameter.
2. Describe **radial** and **tangential** distortion, the OpenCV coefficients `(k1, k2, p1, p2, k3)`, and when you need the **fisheye** or **omnidirectional** model instead.
3. Compare camera types (standard, wide-angle, fisheye, 360°, stereo, RGB-D, rolling vs global shutter) and calibration targets (chessboard, circles, ChArUco).
4. **Calibrate** a camera with OpenCV: corners → object/image points → `calibrateCamera` → **reprojection error**.
5. **Undistort** images and video efficiently, and judge a calibration's quality.
6. Calibrate a **stereo pair** and rectify it; calibrate your own **Raspberry Pi / USB camera**.
7. Apply a **perspective transform** to get a bird's-eye view and measure lane curvature.

Roadmap

① Why? distortion matters → **② Pinhole** K and [R | t] → **③ Distortion** radial & tangential → **④ Camera types** fisheye, 360°, stereo… → **⑤ Calibrate** chessboard → K, dist → **⑥ Beyond** ChArUco, fisheye, stereo, Pi → **⑦ Perspective** bird's-eye & curvature



The whole lecture in one picture: real world → (lens) → distorted image → (calibration) → corrected image (Udacity).

Camera calibration in one minute



1 · Show a known shape

Hold a printed **chessboard** in front of the camera. We know exactly where every corner is on the paper.



2 · See where it lands

Find the corners in the picture. Because of the lens and the viewpoint they are **not** where a perfect camera would put them.



3 · Solve for the numbers

Find the few numbers (**focal length**, **image centre**, **lens distortion**) that explain the difference. Do it once, use it forever.

Analogy: an eye doctor shows you a chart whose letters she knows, sees how you read it, and writes your **prescription**. The chessboard is the chart, the camera is the eye, and **K + dist** is the camera's prescription. Afterwards we "put the glasses on": every image is corrected, and pixels can be turned into millimetres.

Key words in plain English

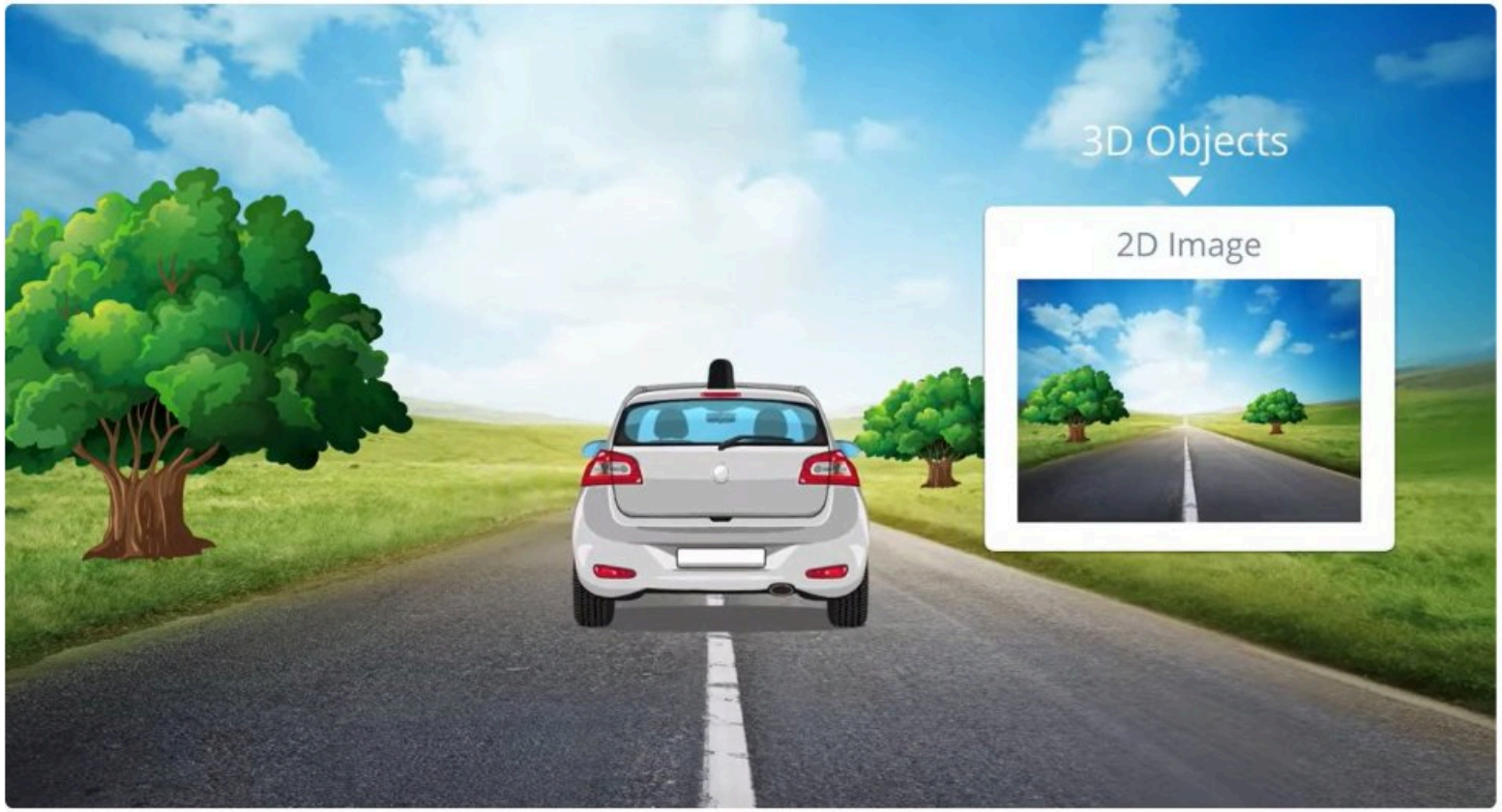
Word	Means	Everyday picture
Intrinsics, K	numbers inside the camera: focal length f_x, f_y and image centre c_x, c_y (in pixels)	the camera's ID card
Extrinsics, [R | t]	where the camera **is** (t) and where it **looks** (R)	your GPS position + which way you face
Distortion, dist	how much the lens bends lines near the edges: k1, k2, k3 (radial), p1, p2 (tangential)	looking through a peephole or a glass of water
Object points	where the board corners **really** are (mm, Z = 0)	the answer sheet
Image points	where the corners **appear** in the photo (pixels)	the student's answers
Reprojection error	distance between where the model predicts a corner and where it was found	how far a dart lands from the bullseye
Undistort	re-draw the image as a perfect pinhole camera would have taken it	putting the glasses on
Homography / perspective transform	a 3 × 3 recipe that re-draws a **flat** thing as seen from another angle	photographing a poster straight-on

01

Why calibrate a camera?

Cameras do not create perfect images

Image distortion: 3-D world → imperfect 2-D image

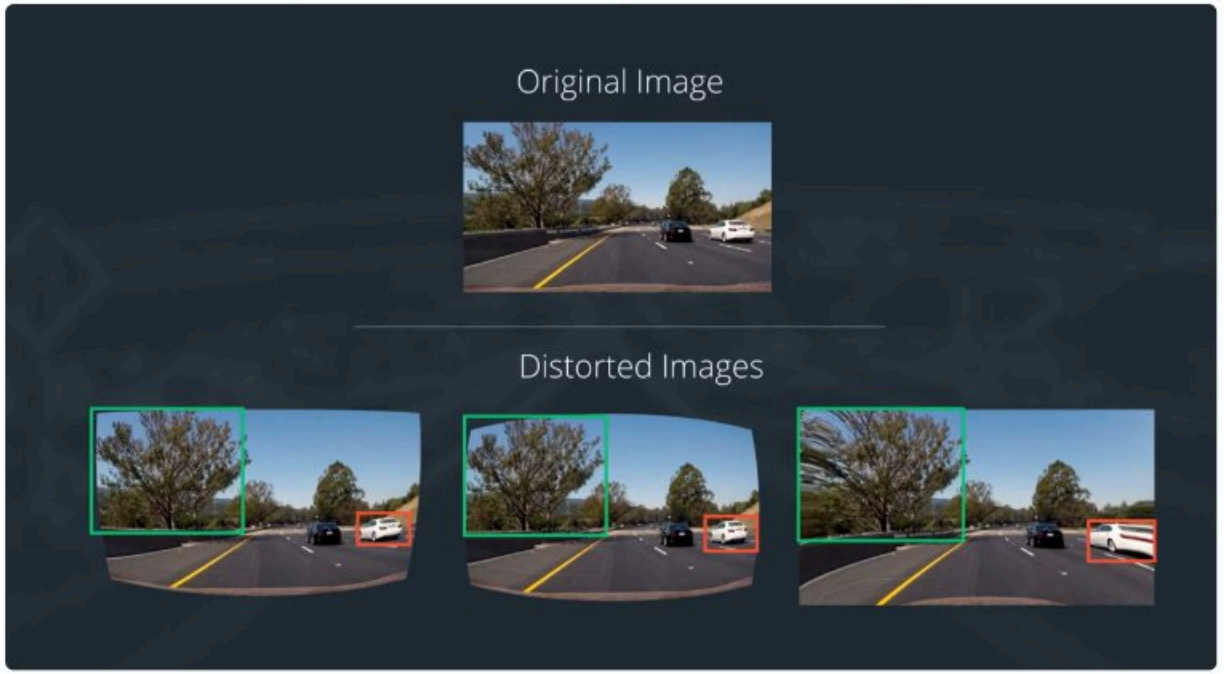


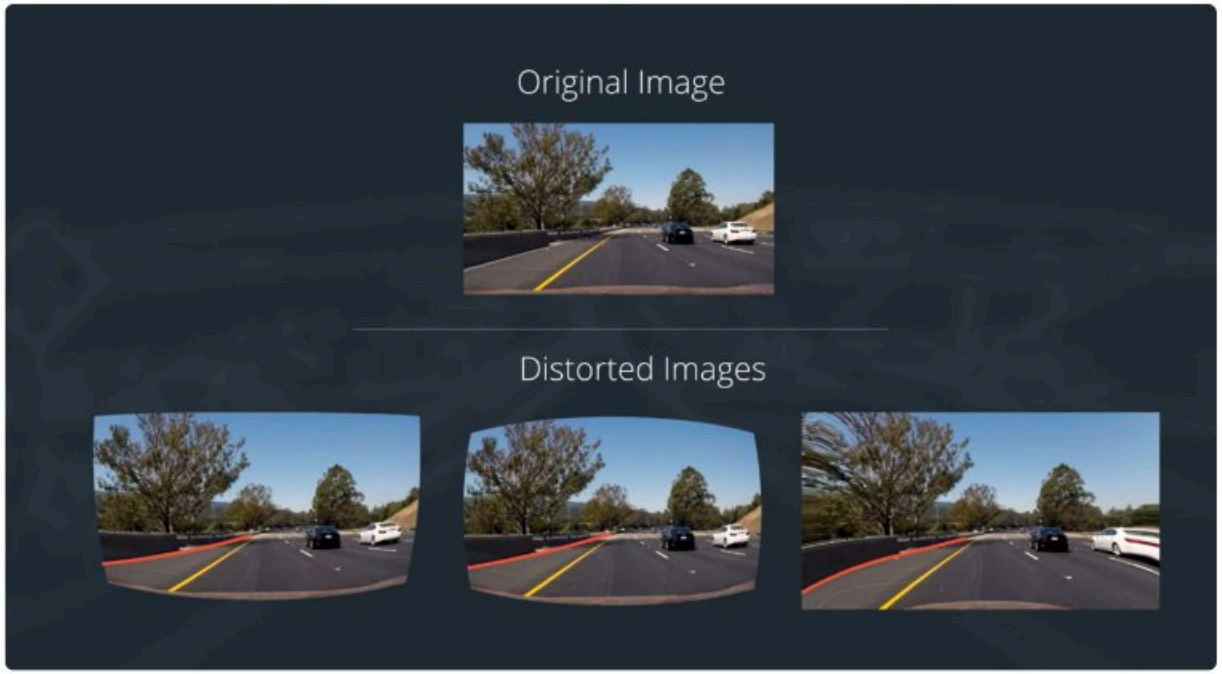
When a camera looks at **3-D objects** and turns them into a **2-D image**, the transformation is not perfect.

- Distortion changes the apparent **shape** and **size** of objects
- It is strongest near the **edges** of the image
- A distorted lane gives the **wrong curvature** → the wrong **steering angle**

> So the **first step** of any measuring vision system is to **undo the distortion.**

The same road through different lenses

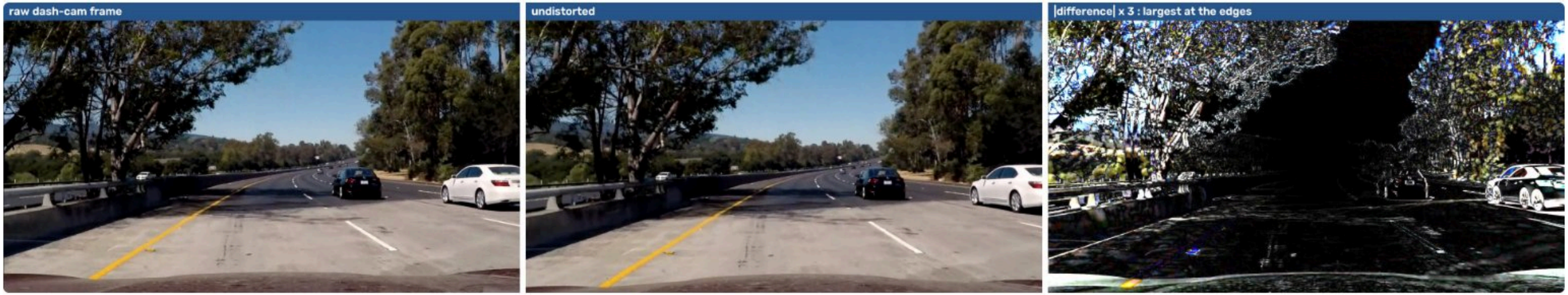




The lane edges are **bent**, rounded or stretched outward. Distortion changes what the shape and size of objects appear to be, and a self-driving car must place itself in the world **accurately**.

Is it visible on a real dash-cam? new

▶ Lab 6D in Colab



A frame from the Udacity Advanced Lane Lines camera before and after undistortion (we calibrate this camera in Lab 6D). The difference image shows the correction is **largest near the image border**: exactly where the lane lines and nearby cars are.

Quiz: why is it important to correct for distortion?

QUIZ QUESTION · SELECT ALL THAT APPLY

a. Distortion can change the apparent **size** of an object in an image.
b. Distortion can change the apparent **shape** of an object in an image.
c. Distortion can cause an object's appearance to change depending on **where it is in the field of view**.
d. Distortion can make objects appear **closer or farther away** than they actually are.

> **All four.** Each one breaks measurements: lane width, curvature, object size, distance to the car ahead.

What calibration gives you

Intrinsics K

Focal lengths f_x, f_y and principal point c_x, c_y, in pixels. Fixed for a given camera, lens, focus and resolution.

Distortion

`dist = [k1, k2, p1, p2, k3]` (or the fisheye `[k1..k4]`). How the lens bends rays away from the ideal pinhole.

Extrinsics [R | t]

Where the camera is relative to the world (or to the board, or to a second camera). One pose per calibration image.

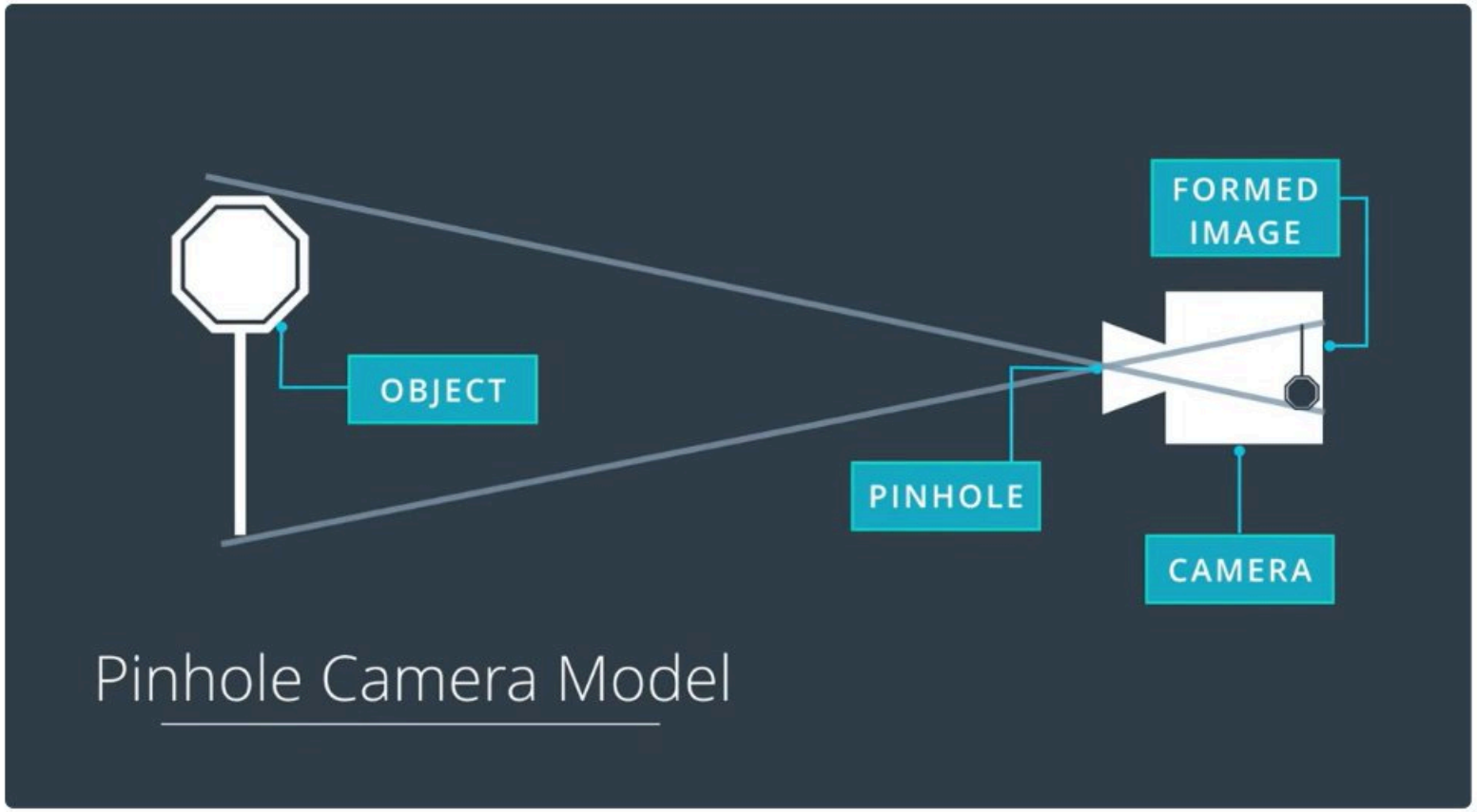
Application	Needs
Lane curvature, bird's-eye view (SDC)	K + distortion, then a perspective transform
Measuring object size in mm (inspection, Lab 9)	K + distortion + the distance or plane
Stereo depth (Lecture 10), visual SLAM, 3-D reconstruction	K, distortion **and** the relative pose of the cameras
AR overlays, robot pick-and-place, ArUco pose (`solvePnP`)	K + distortion → pose of objects
Multi-sensor fusion (camera + LiDAR / IMU)	intrinsics + extrinsic calibration between sensors

02

The pinhole camera model

How 3-D points become pixels

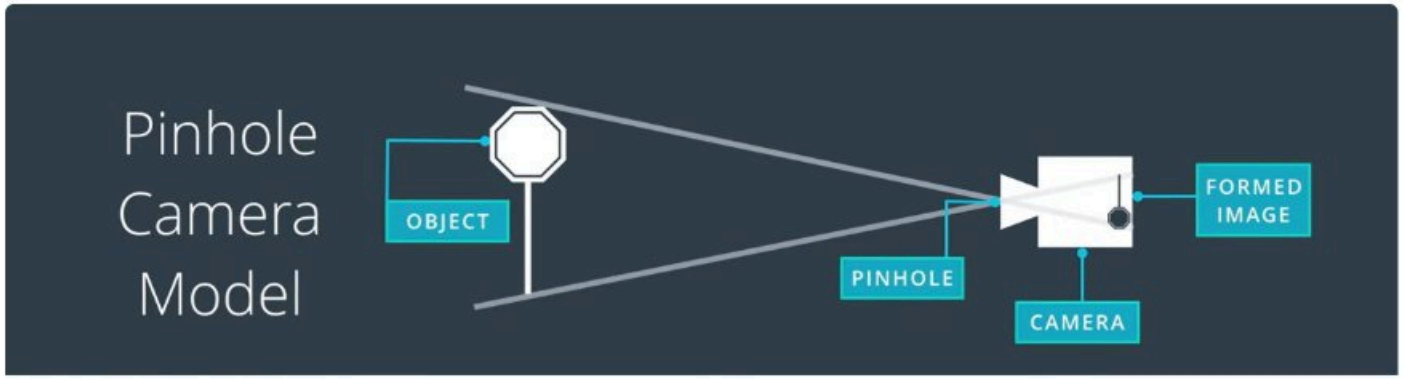
The pinhole camera

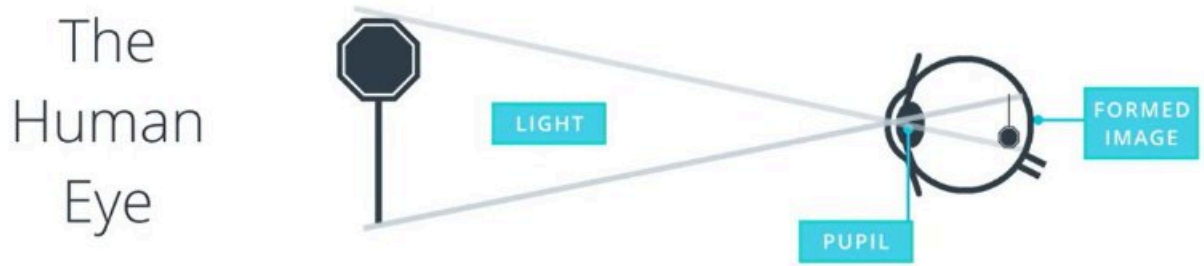


- Light reflected from an object passes through a **tiny hole** and forms an image on the back plane (film / sensor).
- The image is **upside-down and mirrored**: rays from the top of the object continue through the pinhole to the bottom of the image.
- The pinhole camera is the earliest means of projecting images: the camera obscura was described in China around 2,500 years ago.

> A pinhole has **no lens**, so it has **no distortion**; it just lets in very little light.

The camera and the human eye





The eye works the same way: the **pupil** is the pinhole and the **retina** is the image plane, and the image on it is upside-down too. The brain flips it.

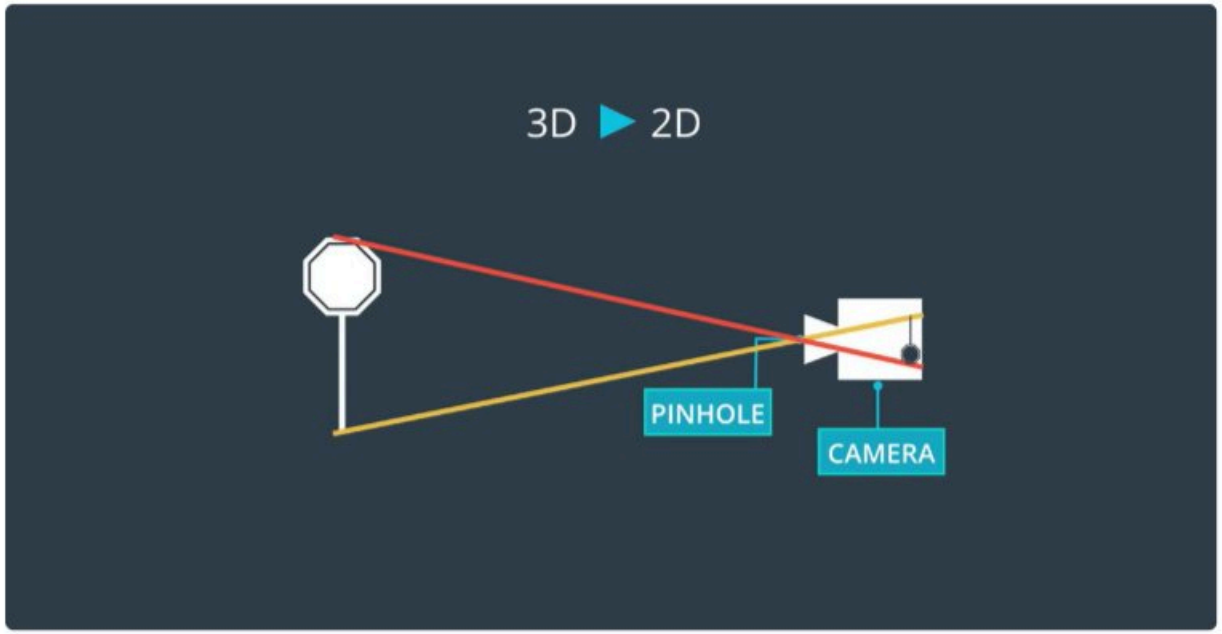
QUIZ

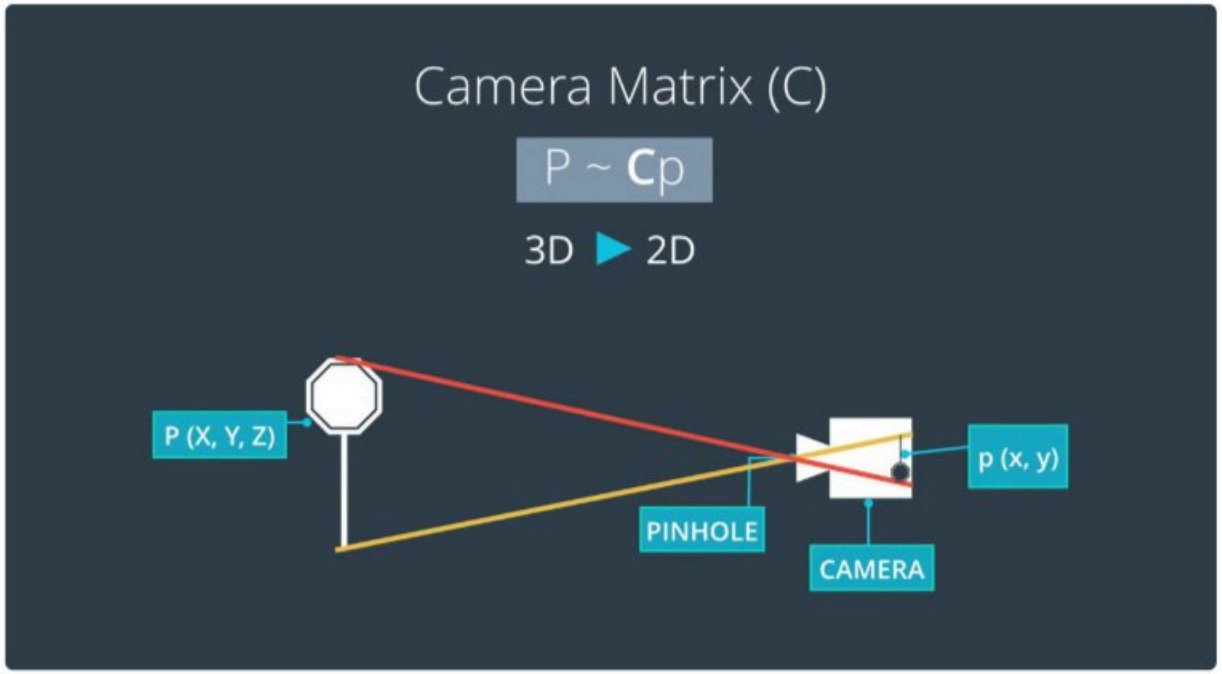
What is the fundamental difference between images formed with a pinhole camera and those formed using lenses?

- Pinhole images are always flipped, lens images upright
- Pinhole images are black and white
- Pinhole images are free from distortion; lenses tend to introduce distortion
- They are exactly the same

> **Free from distortion.** Lenses gather more light but bend rays at their edges.

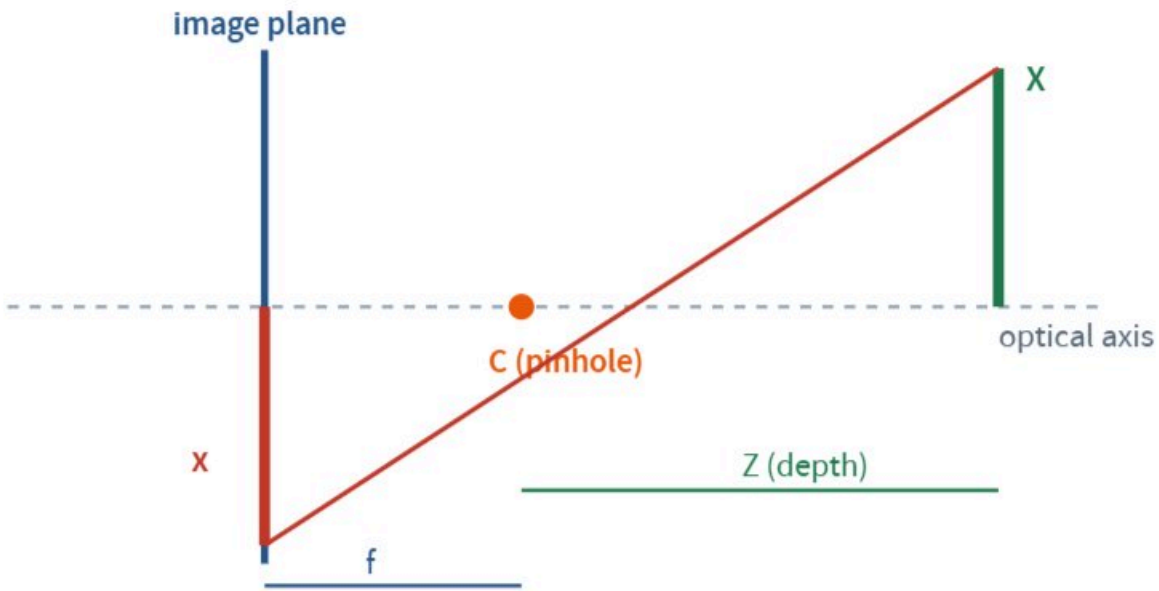
From 3-D to 2-D: the camera matrix





A 3-D object point **P(X, Y, Z)** maps to a 2-D image point **p(x, y)** through a **camera matrix** (Udacity calls it C, so p ≈ C P). Calibration is the job of **measuring** that matrix plus the lens distortion.

Perspective projection by similar triangles



Similar triangles give, for a camera at the origin looking along Z:

$$x = f\,\frac{X}{Z}, \qquad y = f\,\frac{Y}{Z}$$

- Objects **twice as far** appear **half as big**: the **1/Z** is perspective.
- Depth is **lost**: every point on the ray through C lands on the same pixel. One camera cannot measure distance on its own.
- We put a virtual image plane **in front** of the pinhole so the image is not flipped.

In plain words: a camera is a box with a hole. Far things make small pictures (divide by Z), and a single photo cannot tell how far away something is.

From millimetres to pixels: the intrinsic matrix K

$$\begin{bmatrix} u \\ v \\ 1 \end{bmatrix} \simeq \underbrace{\begin{bmatrix} f_x & s & c_x \\ 0 & f_y & c_y \\ 0 & 0 & 1 \end{bmatrix}}_{K} \begin{bmatrix} X/Z \\ Y/Z \\ 1 \end{bmatrix}$$

- **f_x, f_y**: focal length in **pixels** = f [mm] / pixel size [mm]. Equal for square pixels.
- **c_x, c_y**: **principal point**, where the optical axis hits the sensor (≈ image centre, never exactly).
- **s**: skew, 0 for every modern sensor (OpenCV assumes 0).

Worked example: Pi Camera v2

Sony IMX219: f = 3.04 mm, pixel = 1.12 µm, 3280 × 2464 px.

f_x = 3.04 / 0.00112 ≈ **2714 px**

c_x ≈ 3280/2 = **1640**, c_y ≈ **1232**

Horizontal FOV = 2·atan(1640 / 2714) ≈ **62°** ✓ (datasheet: 62.2°)

Datasheet values are only a starting guess. Calibration measures the real ones.

In plain words: K turns "how far left/up the ray points" into a pixel position. f says how much the camera zooms; (c_x, c_y) is the pixel straight in front of the lens.

Try it: K from a datasheet interactive

Pi Cam v2 | Pi Cam 3 | Pi Cam 3 Wide | Global Shutter | USB webcam

focal length f (mm) 3.04

pixel size (µm) 1.12

width (px) 3280 height (px) 2464

resize to width (px) 640 (scaling the image scales K!)

```
full resolution 3280 x 2464
fx = fy = f / pixel = 2714.3 px

K = [[  2714.3,      0.0,   1640.0],
     [     0.0,   2714.3,   1232.0],
     [     0.0,      0.0,      1.0]]

horizontal FOV = 62.3 deg
vertical FOV   = 48.8 deg

resized to 640 x 481  (scale 0.195):
fx' = 529.6   cx' = 320.0   cy' = 240.4
```

Sensor data from the Raspberry Pi camera documentation. Lens values of webcams vary: treat the preset as an example.

Where is the camera? Extrinsics [R | t]

World points are usually not given in the camera's frame. A **rotation R** (3×3) and **translation t** (3×1) move them there:

$$\mathbf{X}_{cam} = R\,\mathbf{X}_{world} + \mathbf{t}$$

Putting it all together, the full **projection** (Lecture 10 notation):

$$s\begin{bmatrix} u \\ v \\ 1 \end{bmatrix} = K\,[R \mid \mathbf{t}]\begin{bmatrix} X \\ Y \\ Z \\ 1 \end{bmatrix} = P\,\tilde{\mathbf{X}}$$

P = K[R | t] is the 3×4 **projection matrix**: 4 intrinsic + 6 extrinsic numbers (3 rotation, 3 translation).

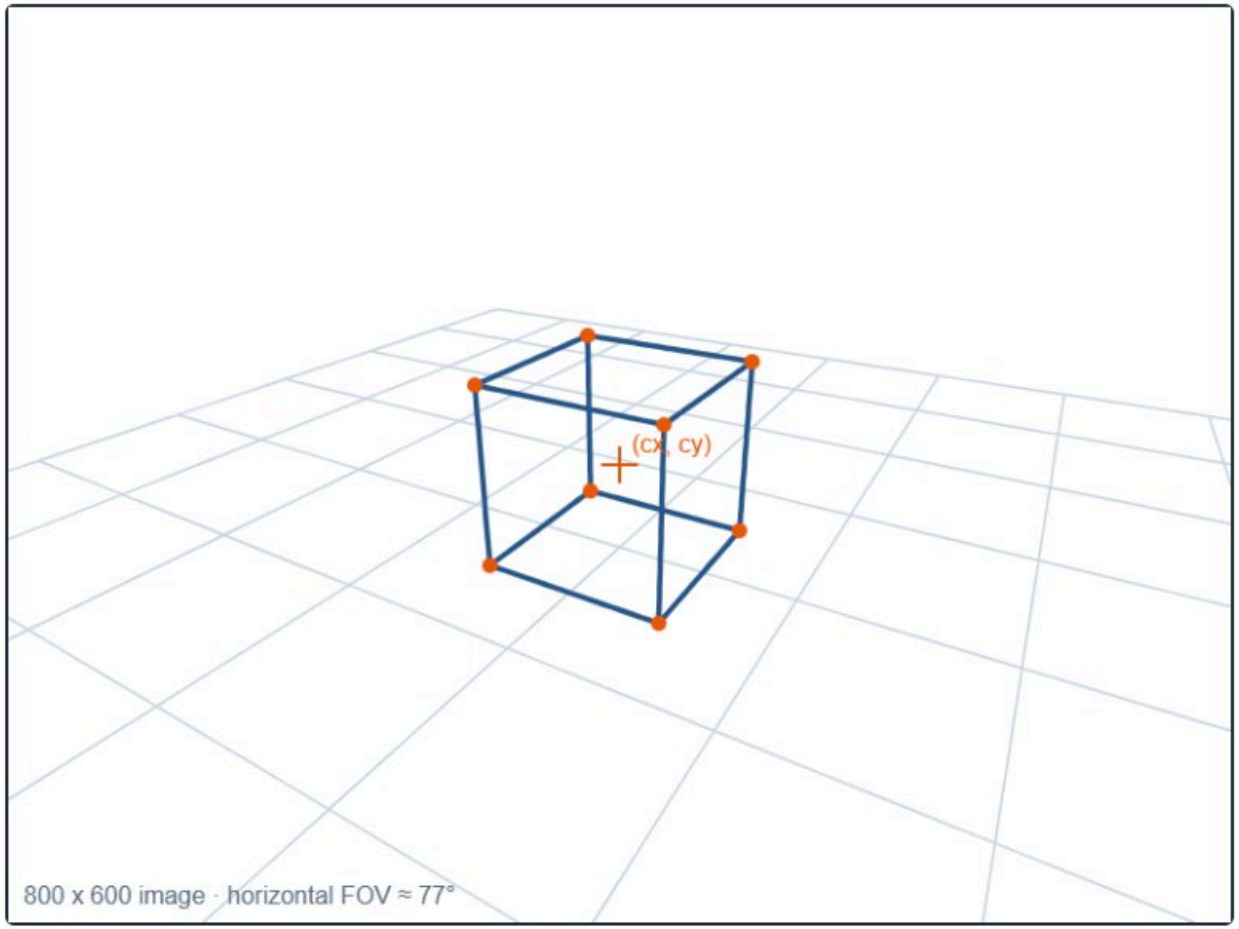
In OpenCV

- R is stored as a 3-vector `rvec` (axis × angle, **Rodrigues**): `R, _ = cv2.Rodrigues(rvec)`
- `calibrateCamera` returns one `rvec, tvec` **per calibration image**: the pose of the board
- `cv2.projectPoints(X, rvec, tvec, K, dist)` applies the whole model, including distortion
- `cv2.solvePnP` does the inverse: known K + 3-D/2-D matches → pose

In plain words: first move the world so the camera is at the origin looking along Z (R, t), then let K turn it into pixels.

Live demo: the pinhole camera interactive

▶ Lab 6A in Colab



focal length f x = f y 500

principal point c x 400

distance Z (m) 4.0

orbit around (°) 30

camera elevation (°) 25

barrel distortion k 1 0.00

```
K = [[  500,     0,  400],
     [    0,   500,  300],
     [    0,     0,    1]]
camera centre C = [1.81, -1.19, -3.14] m
dist = [0.00, 0, 0, 0, 0]
```

A 1 m cube on a floor grid of 1 m squares; the camera looks at the cube. Change f: the image zooms (FOV changes). Move Z: size ∝ 1/Z. Add k1 < 0: straight edges bow outwards.

Quiz: project a point by hand

▶ Lab 6A in Colab

QUIZ QUESTION

A camera has $f_x = f_y = 800$ px, $c_x = 640$, $c_y = 360$, no distortion, and R = I, t = 0. Where does the point $\mathbf{X} = (0.5, -0.25, 4)$ m land?

> $u = 800 \cdot 0.5/4 + 640 = 740$, $v = 800 \cdot (-0.25)/4 + 360 = 310$. Pixel (740, 310).

And a point twice as far, $(1, -0.5, 8)$?

> **The same pixel (740, 310)**: it lies on the same ray. Depth cannot be recovered from one image.

```python
import numpy as np, cv2
K = np.array([[800, 0, 640],
              [0, 800, 360],
              [0,   0,   1]], float)
X = np.array([[0.5, -0.25, 4.0],
              [1.0, -0.50, 8.0]])
uv, _ = cv2.projectPoints(X, np.zeros(3), np.zeros(3),
                          K, None)
print(uv.reshape(-1, 2))   # [[740. 310.] [740. 310.]]
```

Image formation: the full chain in OpenCV

World point
(X, Y, Z) in metres

→

Extrinsics
$X_c = R\,X + t$

→

Normalise
$x' = X_c/Z_c,\ y' = Y_c/Z_c$

→

Distort
radial + tangential
(x'', y'')

→

Intrinsics
$u = f_x x'' + c_x$
$v = f_y y'' + c_y$

Distortion is applied to the **normalised** coordinates (units of focal length), not to pixels. That is why the same `dist` vector works at any image size, while **K must be scaled** when you resize.

Udacity's notes write r as the distance from the **distortion centre** (≈ image centre). In OpenCV it is $r^2 = x'^2 + y'^2$ in normalised coordinates.

Watch: Camera Intrinsics and Extrinsics: 5 Minutes with Cyrill (Cyrill Stachniss, Uni Bonn).

REMEMBER

The pinhole model: the 3 things to remember

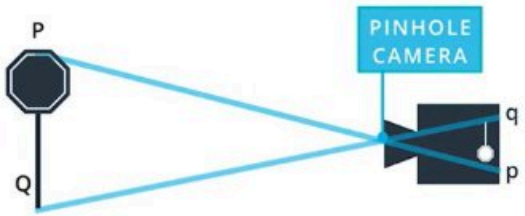
1. A camera divides by depth: **x = f·X/Z**. Far things are small, and depth is lost.
2. **K** (f_x, f_y, c_x, c_y) = the camera itself. **[R | t]** = where it stands and looks.
3. In OpenCV: `cv2.projectPoints(X, rvec, tvec, K, dist)` does the whole chain.

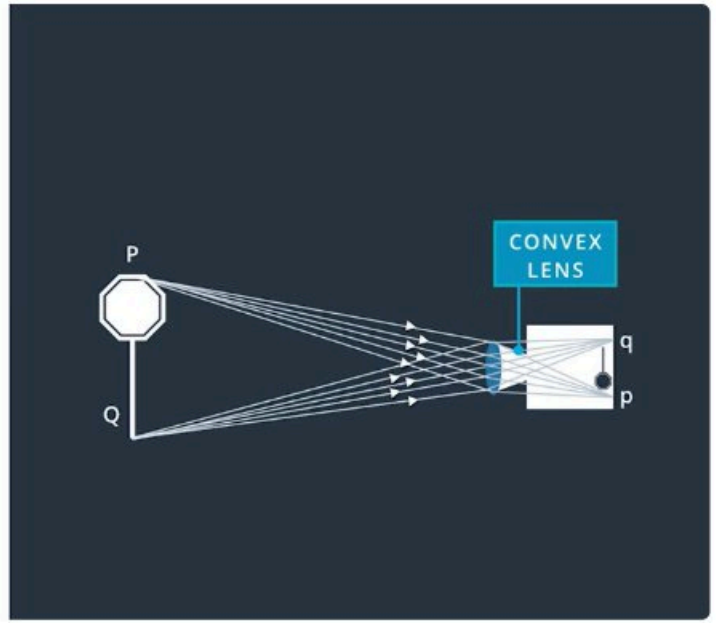
03

Lens distortion

Real cameras use lenses, and lenses bend light

Why real cameras use lenses





- A pinhole lets through **one ray** per scene point: a dark image or a very long exposure.
- A **convex lens** focuses **many rays** at once: bright images and fast shutter speeds.
- But rays bend a little **too much or too little** near the edges of a curved lens → **distortion**.

Radial distortion: the most common type

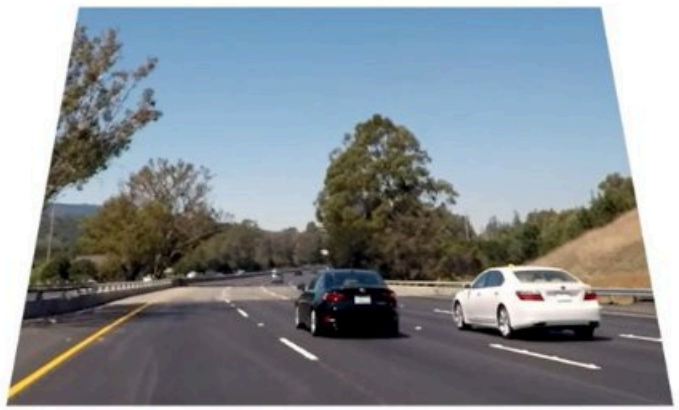


Light rays bend more (or less) the **further they are from the centre** of the lens. Lines and objects appear more or less **curved** than they are.

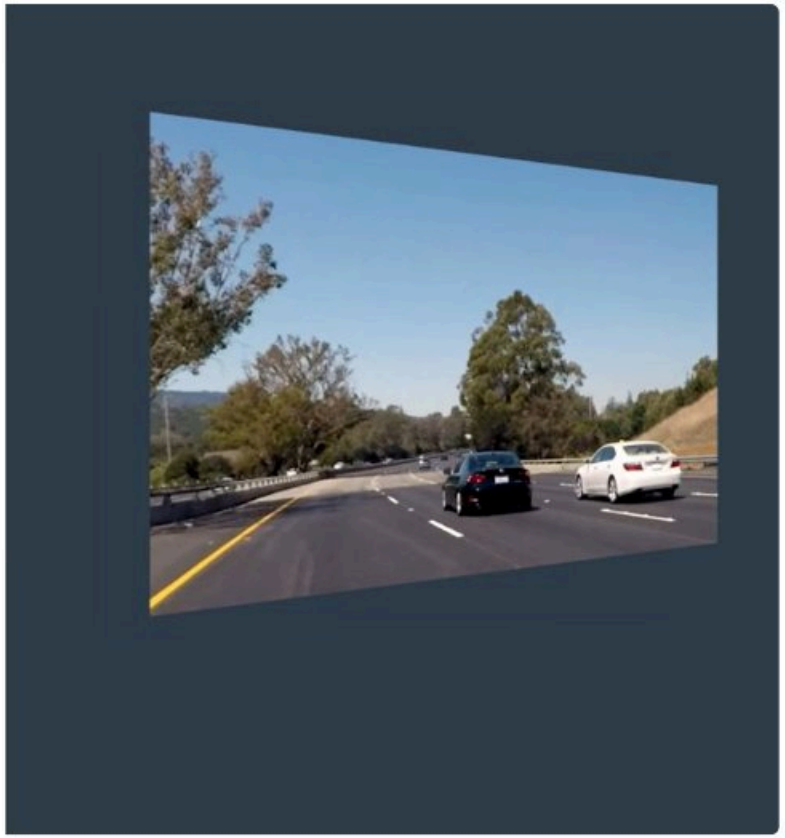
- **Barrel** ($k_1 < 0$): lines bow **outwards**: wide-angle, GoPro, dash-cams
- **Pincushion** ($k_1 > 0$): lines bow **inwards**: telephoto, some zooms
- **Mustache**: barrel in the centre, pincushion at the edge (k_1, k_2 opposite signs)

In plain words: the lens squeezes (barrel) or stretches (pincushion) the picture more and more the further you go from the centre.

Tangential distortion: a tilted lens



Tangential Distortion



Occurs when the lens is **not perfectly parallel** to the imaging plane (sensor), for example from glue or assembly tolerances in cheap modules.

The image looks **tilted**: some objects appear farther away or closer than they actually are.

> Usually much smaller than radial distortion, but cheap phone, Pi and webcam modules can show it.

Some lenses distort on purpose



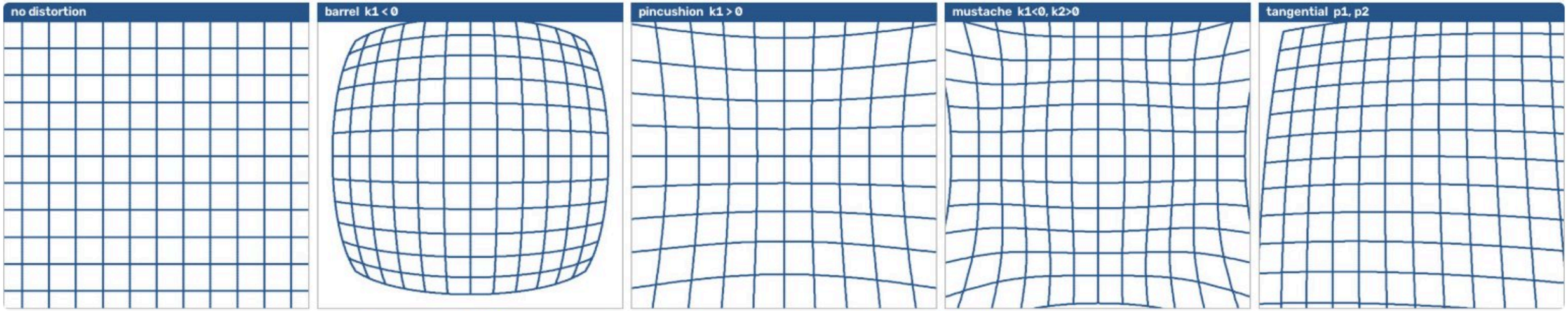
A convex car mirror: wider view, objects look farther away



Fisheye photography: 180° field of view, extreme barrel

Fine for style or a wide view, but for **measurement** (steering, sizing, depth) we need images that accurately reflect the real world.

Distortion types on a straight grid



Generated with `cv2.projectPoints` on a flat grid: same K, different `dist` vectors.

Five numbers capture the distortion

▶ Lab 6A in Colab

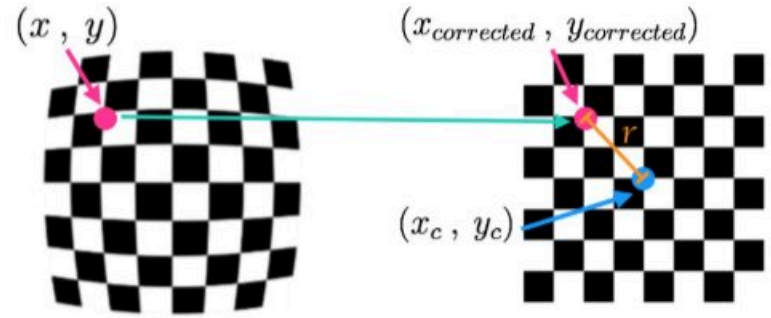
$$\text{Distortion}_{\text{coefficients}} = (k_1\ k_2\ p_1\ p_2\ k_3)$$

Radial, with $r^2 = x^2 + y^2$ (normalised coordinates):

$$x_d = x\,(1 + k_1 r^2 + k_2 r^4 + k_3 r^6)$$
$$y_d = y\,(1 + k_1 r^2 + k_2 r^4 + k_3 r^6)$$

Tangential:

$$x_d = x + [\,2p_1 xy + p_2(r^2 + 2x^2)\,]$$
$$y_d = y + [\,p_1(r^2 + 2y^2) + 2p_2 xy\,]$$



- OpenCV order: `[k1, k2, p1, p2, k3]`. k3 comes **last** because it is only needed for strong (wide-angle) distortion.
- Optional extras: rational `k4–k6`, thin-prism `s1–s4`, tilted sensor `τx, τy`, so up to 14 numbers.
- The equations go **ideal → distorted**. Undistortion inverts them numerically.

Distortion with real numbers (you can do this by hand)

A point near the edge

Normalised point **(x, y) = (0.8, 0.0)**, so $r^2 = 0.64$. Lens: **k1 = −0.2**, all others 0 (a mild barrel).

factor = $1 + k1 \cdot r^2 = 1 - 0.2 \times 0.64 =$ **0.872**

$x_d = 0.8 \times 0.872 =$ **0.698**: the point moves **13 % towards the centre**.

With $f_x = 800$ px that is $(0.8 - 0.698) \times 800 \approx$ **82 pixels**!

A point near the centre

(x, y) = (0.1, 0.0), so $r^2 = 0.01$.

factor = $1 - 0.2 \times 0.01 =$ **0.998**

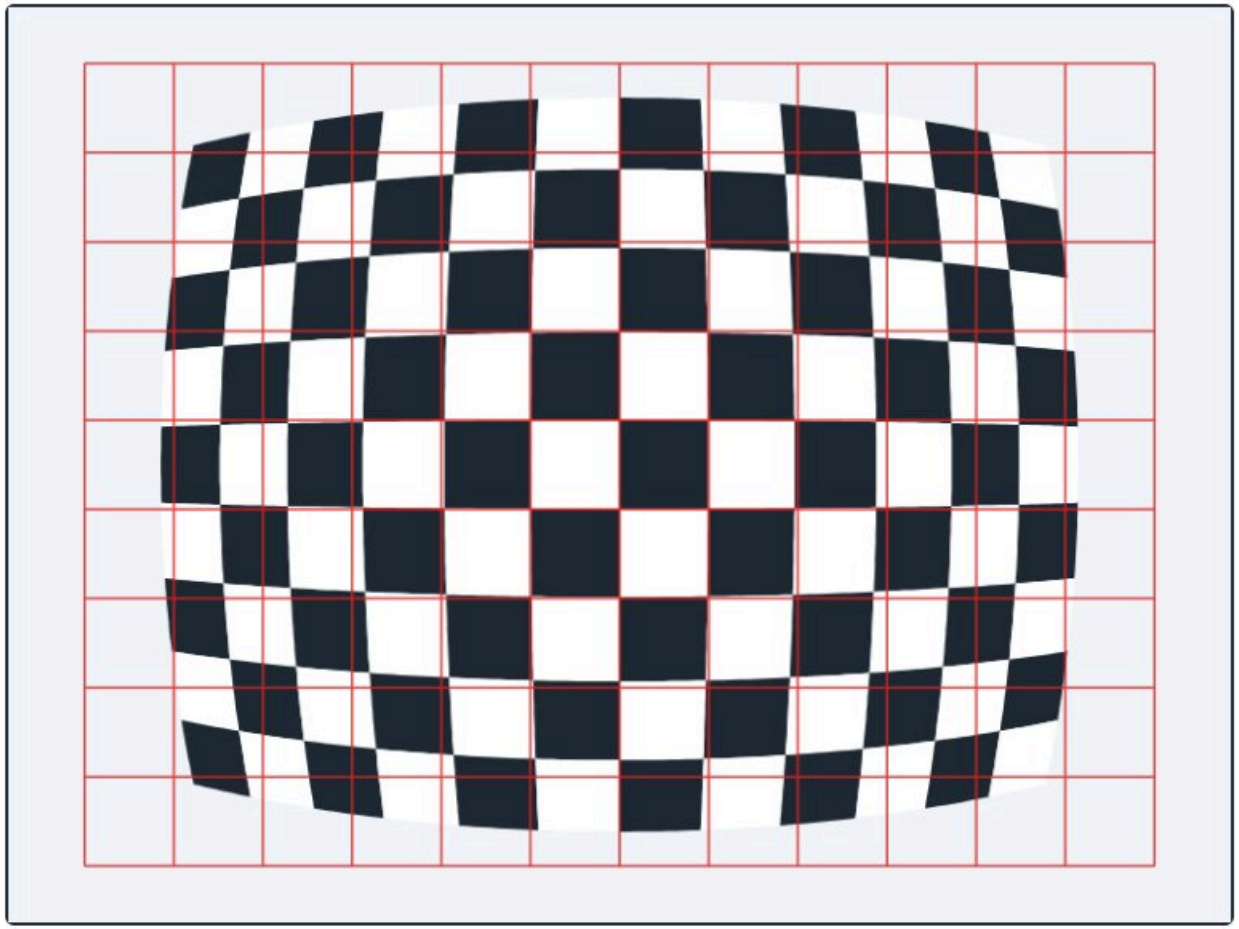
$x_d = 0.0998$: it moves only **0.0002**, about **0.2 pixels**.

That is why the centre of a photo looks fine and the edges bend: the effect grows with r^2.

Take-away: distortion is tiny in the middle and huge at the edges, so calibration photos **must** have the board near the edges and corners.

Live demo: lens distortion explorer interactive

▶ Lab 6A in Colab



k 1 -0.25

k 2 0.05

k 3 0.00

p 1 0.000

p 2 0.000

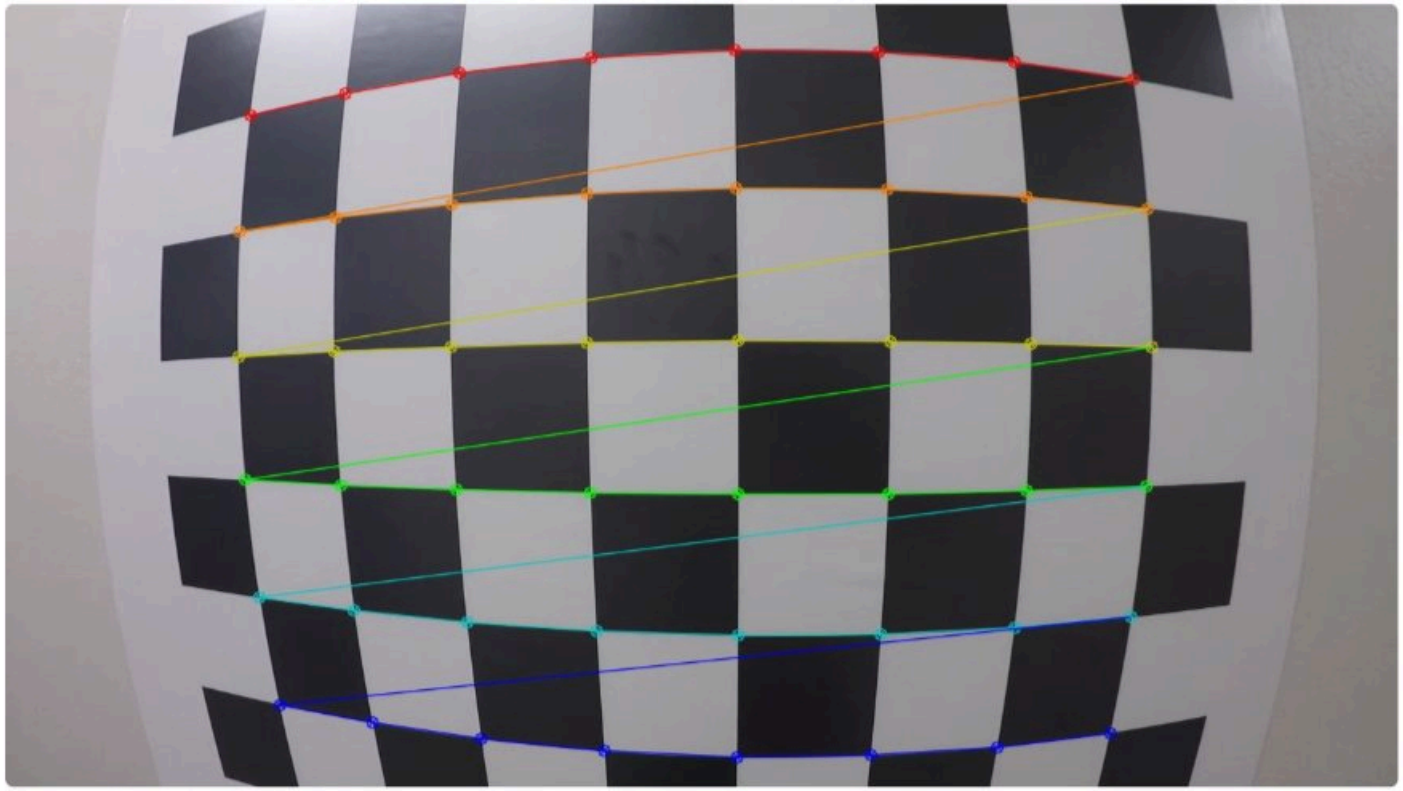
☑ show the ideal (undistorted) board in red

barrel | pincushion | mustache | tangential

GoPro (Lab 6) | reset

```
dist = [k1, k2, p1, p2, k3]
     = [-0.25, 0.05, 0.000, 0.000, 0.00]
```

Quiz: name that distortion



QUIZ QUESTION

This chessboard was shot with a GoPro (the camera we calibrate in Lab 6).

a. What kind of distortion is it? What sign do you expect for $\mathbf{k_1}$?

b. Where in the image is the distortion strongest?

c. Would a pure pinhole camera show it?

> **A:** barrel → $\mathbf{k_1 < 0}$ (Lab 6 measures $k_1 \approx -0.23$). **B:** at the edges and corners (large r). **C:** No, a pinhole has no lens.

REMEMBER

Lens distortion: the 3 things to remember

1. Real lenses bend lines, mostly **radially**: barrel ($k1 < 0$) or pincushion ($k1 > 0$).
2. OpenCV describes it with 5 numbers: `[k1, k2, p1, p2, k3]`.
3. It is small in the centre and large at the edges (it grows with r^2).

04

Types of cameras & their models

One calibration idea, many lens and sensor models

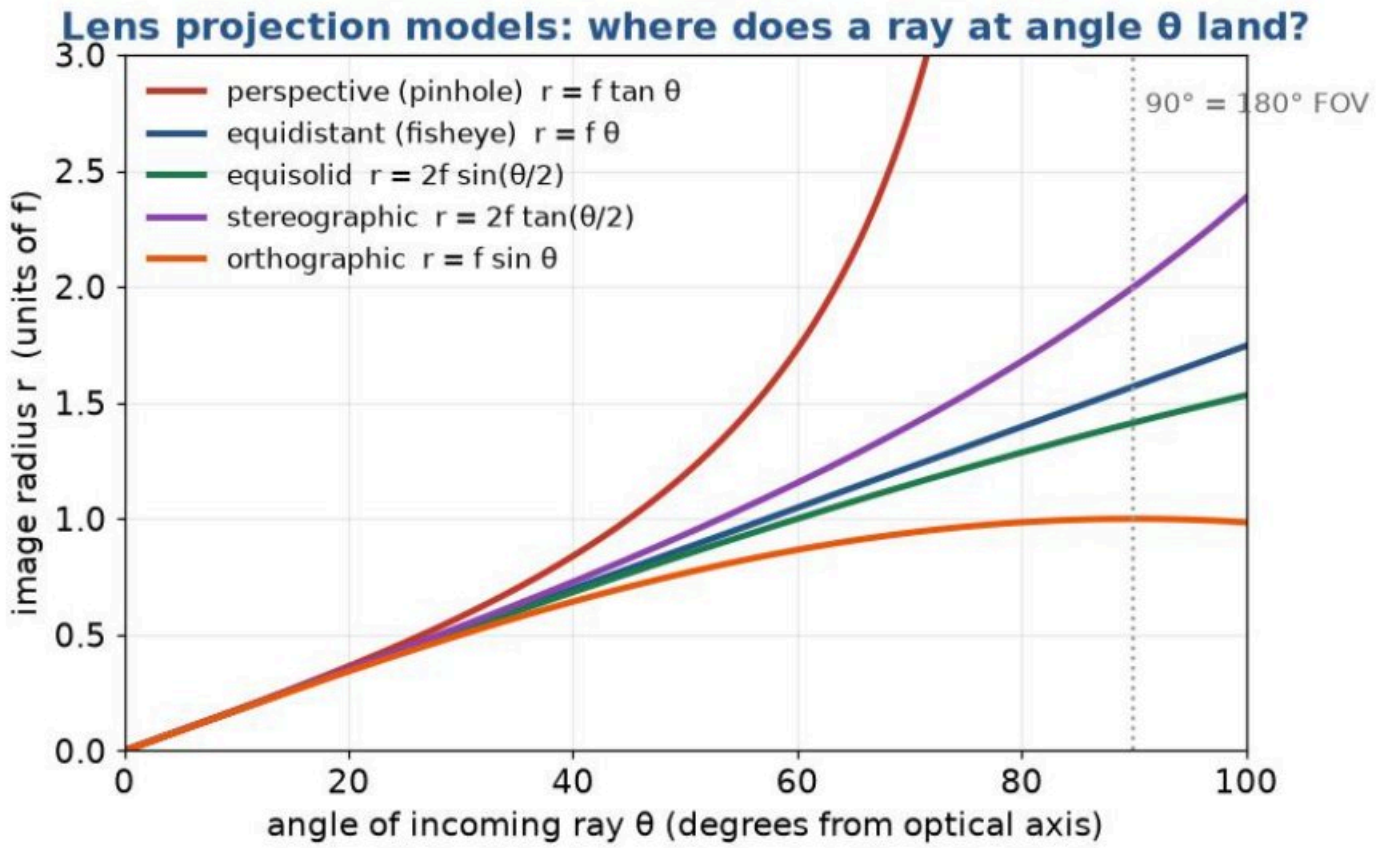
Camera types you will meet

Camera	Typical FOV	Model to calibrate with	OpenCV	Examples
Standard / narrow lens	40–70°	pinhole + radial/tangential (k1, k2, p1, p2)	`cv2.calibrateCamera`	Pi Cam v2/3, webcams, phone main camera
Wide-angle	90–120°	pinhole + k3, or `CALIB_RATIONAL_MODEL` (k4–k6)	`cv2.calibrateCamera`	Pi Cam 3 Wide, GoPro (linear mode), dash-cams
Fisheye	150–200°	Kannala–Brandt equidistant $\theta(1 + k1\theta^2 + \ldots)$	`cv2.fisheye.calibrate`	car surround-view, drones, robot vacuum, T265
Omnidirectional / catadioptric	≥ 180–360°	unified (Mei) model with mirror parameter ξ	`cv2.omnidir.calibrate` (contrib)	mirror + lens rigs, 360° cameras (dual fisheye)
Stereo pair	any	two intrinsics + relative pose (R, T)	`cv2.stereoCalibrate`	ZED, OAK-D, two Pi cameras, Kaggle stereo set
RGB-D / ToF	60–90°	factory-calibrated colour + depth intrinsics & extrinsics	vendor SDK, `calibrateCamera` on IR	RealSense, Kinect, Orbbec
Thermal	25–60°	pinhole: needs a **heated** or emissive target	`calibrateCamera`	FLIR Lepton
Event camera	~ 60–90°	pinhole; board shown as a blinking screen	reconstruct frames → OpenCV	Prophesee, DAVIS

The more the field of view grows, the more the simple pinhole + polynomial model breaks down: at 90° off-axis, $\tan\theta \to \infty$.

> **In plain words:** the wider the lens sees, the more it bends lines, and the more special the maths. Normal lens → standard model; very wide → fisheye model.

Why fisheye needs a different model



- A pinhole maps a ray at angle θ to radius **r = f tan θ**, which goes to infinity at 90°. No finite sensor can hold a 180° view.
- Fisheye lenses are designed so **r grows slowly**: equidistant r = fθ, equisolid, stereographic…
- OpenCV's **fisheye** module models the real lens as $\theta_d = \theta\,(1 + k_1\theta^2 + k_2\theta^4 + k_3\theta^6 + k_4\theta^8)$, so D has **4** coefficients.
- Rule of thumb: FOV above ~120°, or the standard model leaves the edges curved → switch to `cv2.fisheye`.

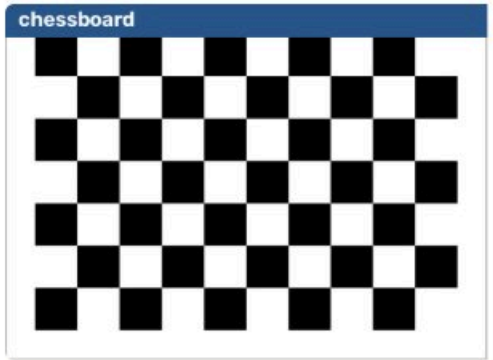
The cameras on our lab bench: Raspberry Pi

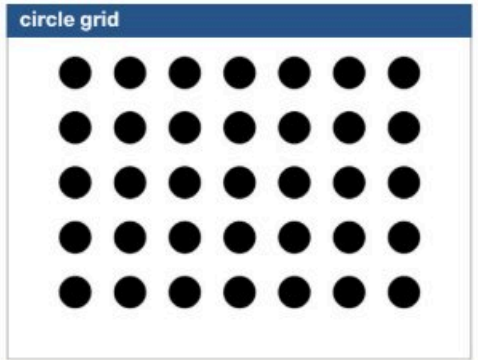
Module	Sensor	Resolution	Pixel	Focal length	H-FOV	Shutter	Calibrate as
Camera Module v2	IMX219	3280 × 2464	1.12 µm	3.04 mm	62°	rolling	standard
Camera Module 3	IMX708	4608 × 2592	1.4 µm	4.74 mm	66°	rolling	standard (fix the focus!)
Camera Module 3 Wide	IMX708	4608 × 2592	1.4 µm	2.75 mm	102°	rolling	wide: k3 or rational
HQ Camera	IMX477	4056 × 3040	1.55 µm	lens (6 mm, 16 mm…)	lens	rolling	depends on lens
Global Shutter Camera	IMX296	1456 × 1088	3.45 µm	lens	lens	**global**	depends on lens
M12 fisheye lens on HQ/GS	–	–	–	~1.5–2 mm	> 180°	–	**fisheye**

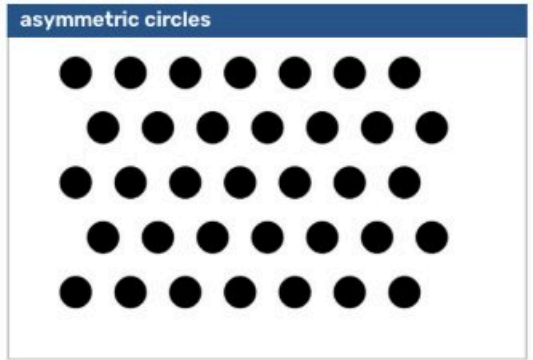
Camera Module 3 has autofocus. Intrinsics change with focus, so lock the lens (`picam2.set_controls({"AfMode": 0, "LensPosition": …})`) before you calibrate and keep it there.

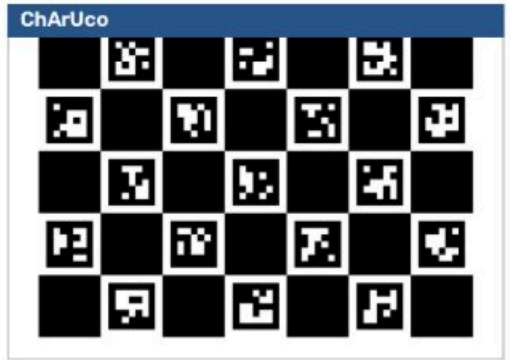
Rolling shutter rows are exposed one after another, so a moving board or camera skews the image. Hold the board **still** for every shot, or use the Global Shutter camera.

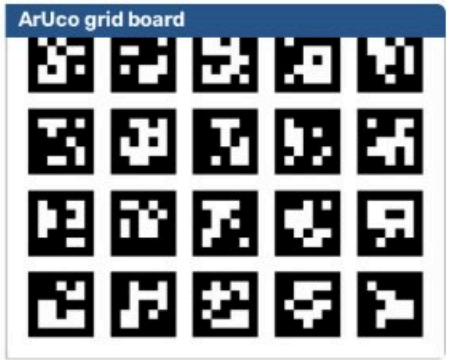
Calibration targets











Target	Detect with	+	–
Chessboard	`findChessboardCorners`, `findChessboardCornersSB`	easy to print, very accurate saddle points	must be **fully visible**; 180° ambiguity if rows = cols
(Asymmetric) circle grid	`findCirclesGrid`	robust to blur & noise	circle centres biased under strong perspective/distortion
ChArUco	`cv2.aruco.CharucoDetector`	**partial views OK**, every corner has an **ID**: ideal for fisheye & edges	needs the right dictionary & a sharp print
ArUco grid / AprilGrid	`cv2.aruco.ArucoDetector`	pose of each marker, multi-camera rigs (Kalibr)	marker corners less precise than saddles

Making a good target (it matters more than the code)

Do

- Print at **100 % scale**, then **measure** a square with a ruler or caliper: that is `square_size`
- Glue it to something **rigid and flat**: glass, aluminium, foam board
- **Matte** paper: no glare spots on the corners
- Keep a **white border** around the pattern (the detector needs it)
- Use **different** row/column counts (9 × 6, not 8 × 8) so the orientation is unambiguous

Don't

- Use a board that bends or ripples: the model assumes **Z = 0** for every corner
- Show it on a laptop screen at an angle (moiré, refresh, reflections). OK only as a last resort
- Count **squares** instead of **inner corners**
- Let auto-exposure / motion blur smear the corners

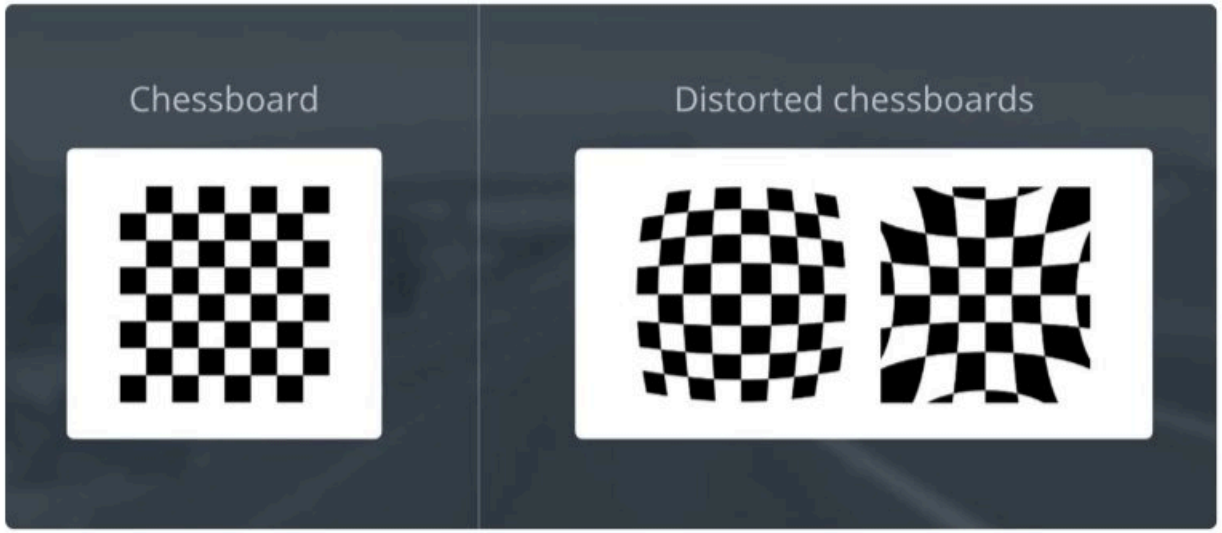
Pattern generators: calib.io, OpenCV `gen_pattern.py` , or `board.generateImage()` in Lab 6E.

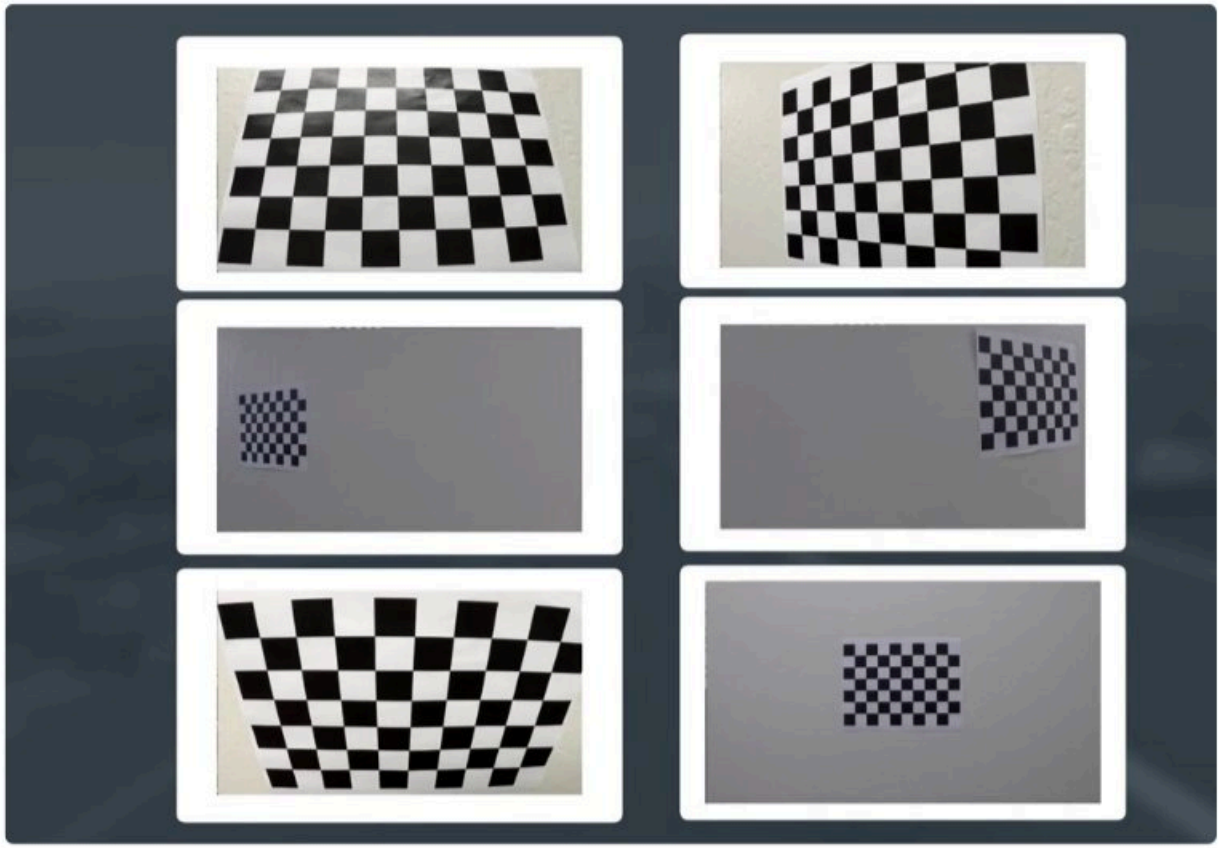
05

Calibrating a camera with OpenCV

Chessboard → corners → object & image points → K, dist

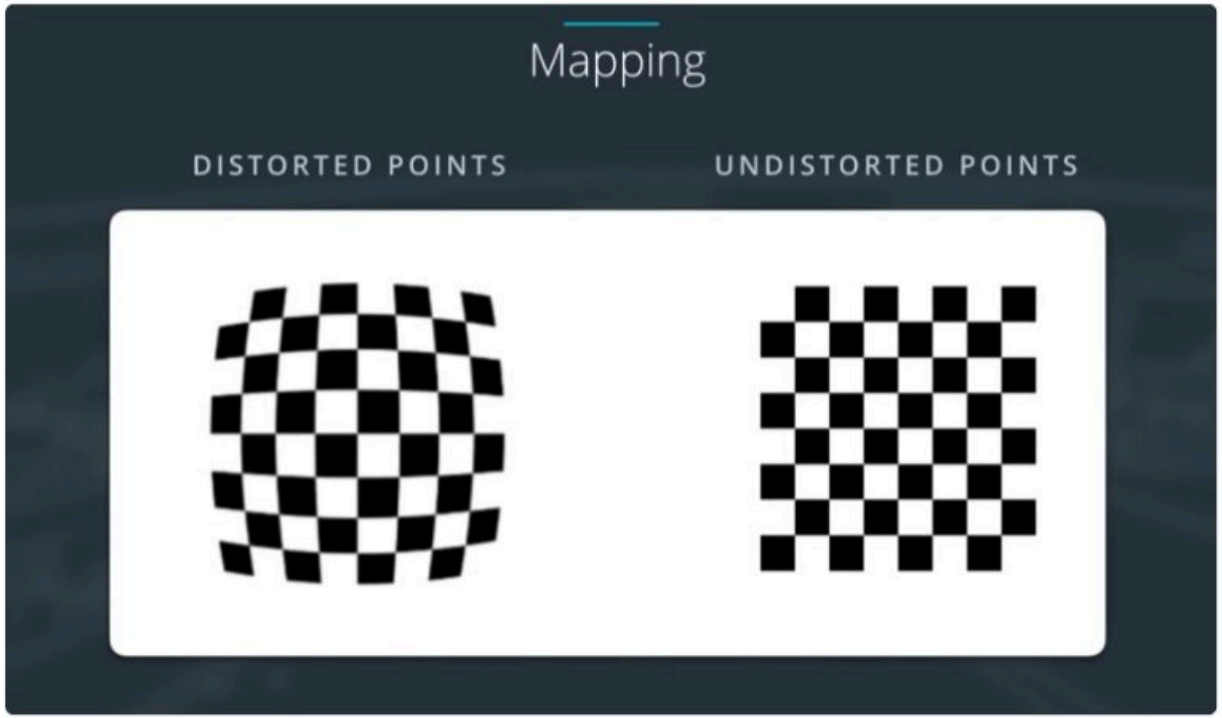
Measuring distortion with a known shape

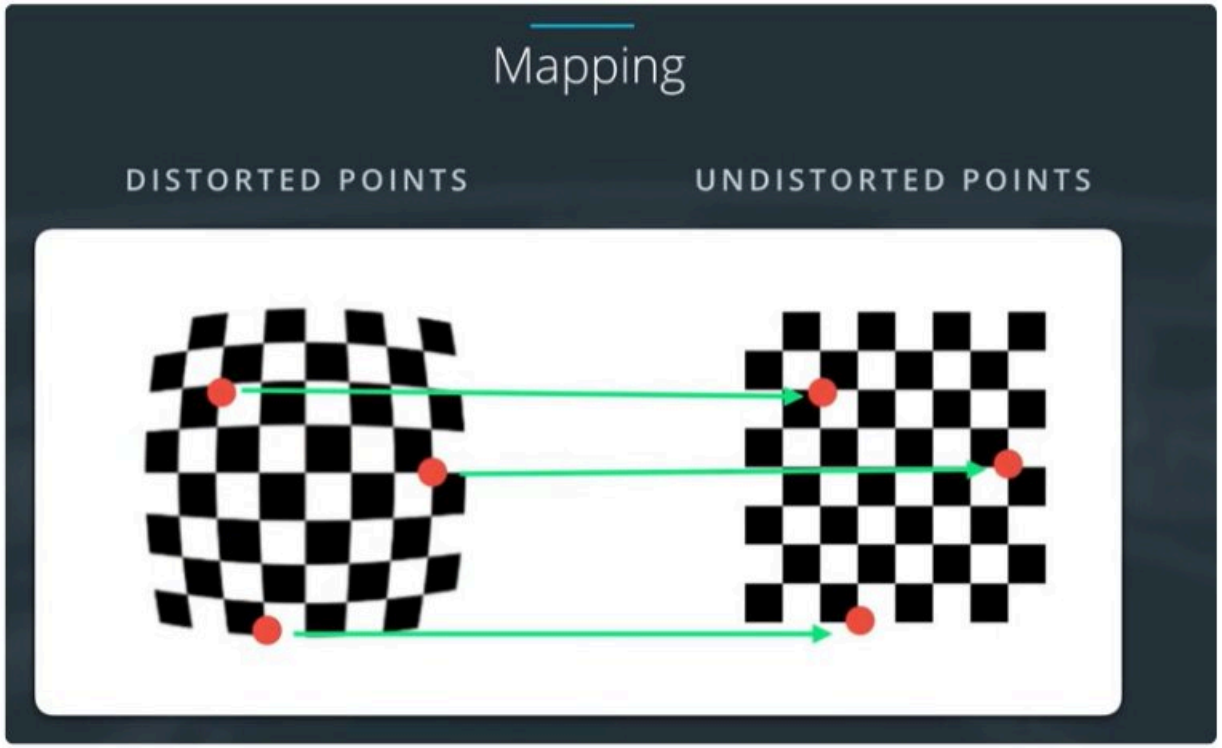




We know exactly what a **flat chessboard** looks like, and its regular high-contrast pattern is easy to detect automatically. Take **many pictures** of it at different angles and distances; the difference between the **apparent** and the **real** square shapes reveals the distortion.

Map distorted points → undistorted points





1. **Detect** the corners in each distorted image: the **image points** (2-D, pixels).
2. Write down where they **really** are on the flat board: the **object points** (3-D, Z = 0).
3. **Calibrate**: find K, dist and each pose that best maps object points to image points.
4. **Undistort** any new image from that camera.

Quiz: how many corners? interactive

▶ Lab 6B in Colab



QUIZ QUESTION

Count the **inside corners** in a row (**nx**) and in a column (**ny**) of the right-hand board. Only points where **two black and two white** squares meet count, not the outside corners.

nx = ____ ny = ____ check

nx = 8, ny = 6. The board has 9 × 7 squares → 8 × 6 inner corners. This is the `patternSize` you pass to OpenCV.

Finding corners: `findChessboardCorners`

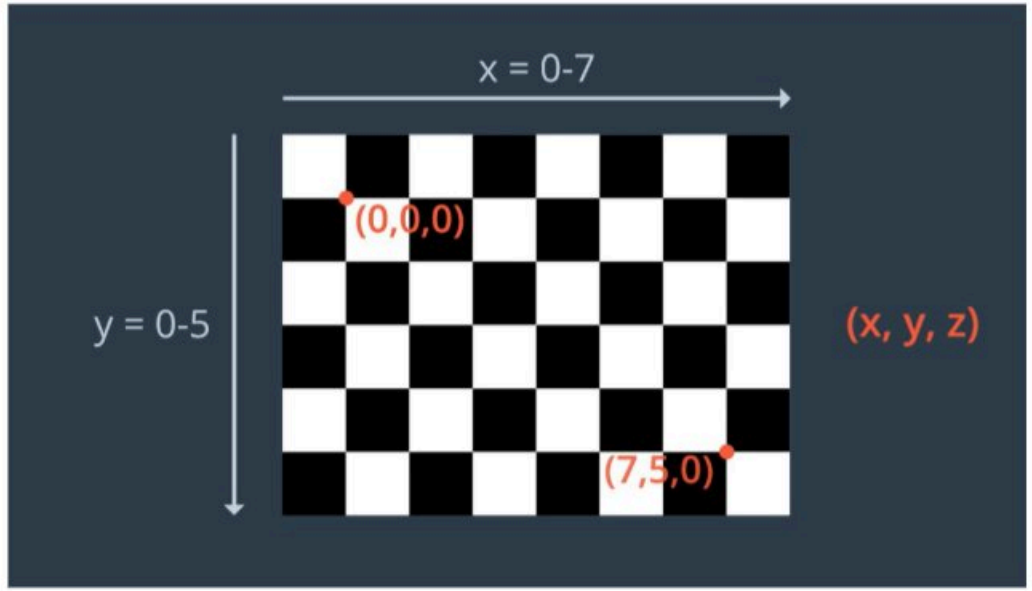
▶ Lab 6B in Colab




```
gray = cv2.cvtColor(img, cv2.COLOR_BGR2GRAY)                # cv2.imread gives BGR (mpimg.imread gives RGB!)
ret, corners = cv2.findChessboardCorners(gray, (8, 6), None)
img = cv2.drawChessboardCorners(img, (8, 6), corners, ret)  # coloured rows; first corner = (0,0,0)
```

Object points: where the corners really are

▶ Lab 6B in Colab



Top-left corner is **(0, 0, 0)**, bottom-right is **(7, 5, 0)**, not (8, 6, 0): counting starts at zero. **Z = 0** for every point because the board is flat.

```python
nx, ny = 8, 6
square = 25.0                # mm, measured on YOUR print

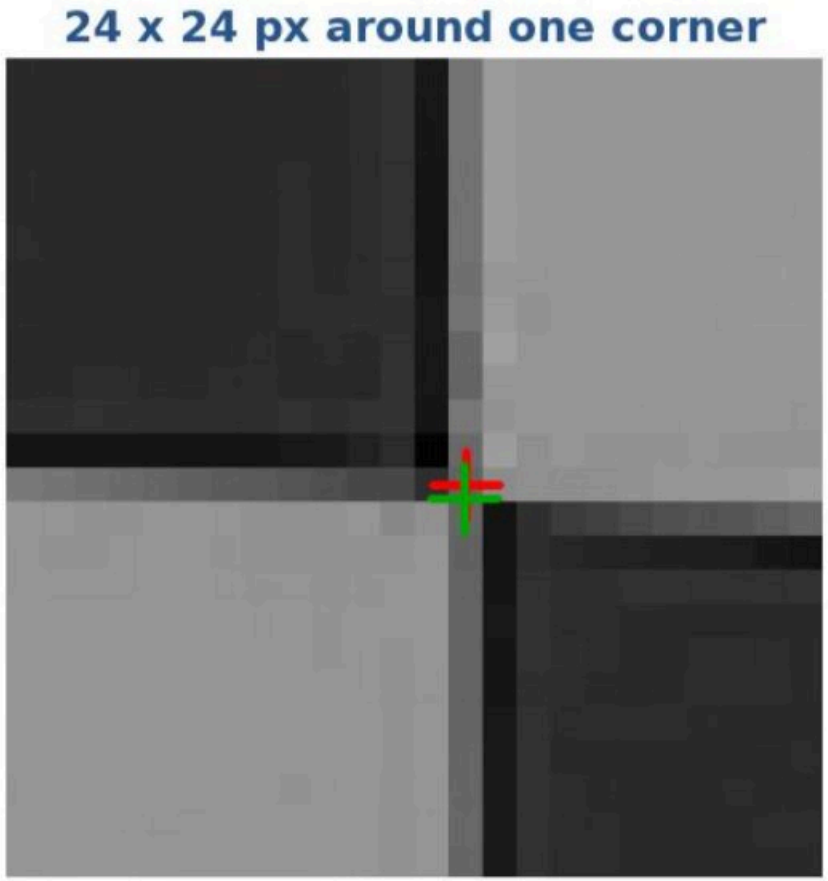
# (48, 3) array: (0,0,0), (1,0,0) ... (7,5,0)
objp = np.zeros((nx * ny, 3), np.float32)
objp[:, :2] = np.mgrid[0:nx, 0:ny].T.reshape(-1, 2)
objp *= square               # now in millimetres

objpoints = []   # 3-D points in the board frame
imgpoints = []   # 2-D points in the image
```

> Units of `objp` = units of `tvec`. In **squares** K and dist are still correct, but distances come out in squares. Multiply by the square size to get mm.

Refining to sub-pixel accuracy

▶ Lab 6B in Colab



+ red: whole-pixel corner + green: cornerSubPix

The saddle point between four squares is not on a pixel centre. `cornerSubPix` moves each corner to where the image gradients around it are orthogonal to the vector to the corner, typically to **0.1 px**.

```
crit = (cv2.TERM_CRITERIA_EPS +
        cv2.TERM_CRITERIA_MAX_ITER, 30, 1e-3)
corners = cv2.cornerSubPix(gray, corners,
                           (11, 11), (-1, -1), crit)

# OpenCV ≥ 4.3: sector-based detector, more
# accurate and robust, sub-pixel built in
ret, corners = cv2.findChessboardCornersSB(
    gray, (8, 6), cv2.CALIB_CB_EXHAUSTIVE +
                  cv2.CALIB_CB_ACCURACY)
```

Window (11, 11) means a 23 × 23 search area: keep it smaller than one square in the image.

Calibration, step 1: collect points from every image

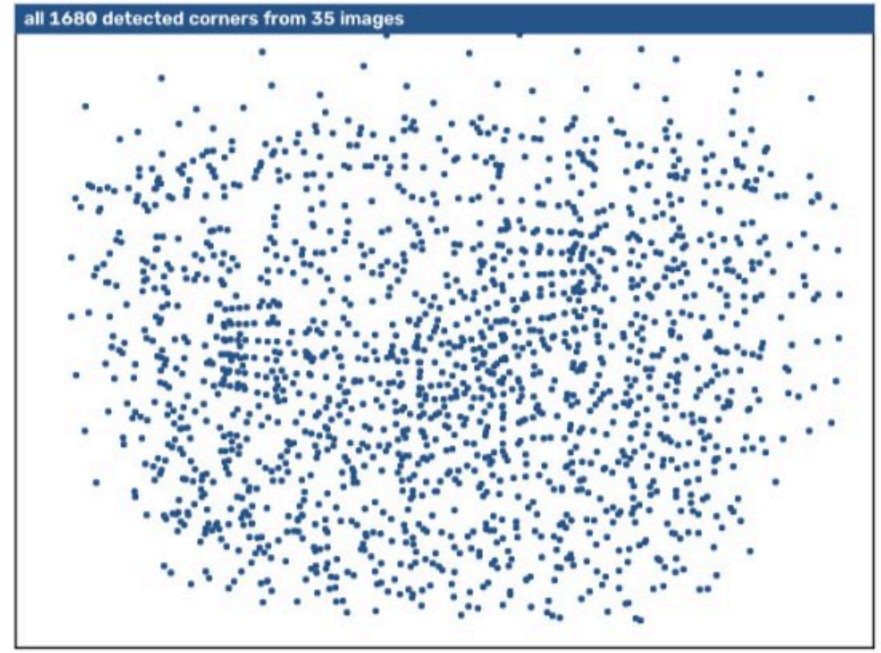
```python
import glob, cv2, numpy as np

nx, ny = 8, 6
objp = np.zeros((nx * ny, 3), np.float32)
objp[:, :2] = np.mgrid[0:nx, 0:ny].T.reshape(-1, 2)

objpoints, imgpoints = [], []
images = sorted(glob.glob("calibration_wide/GOPR*.jpg"))

for fname in images:
    img  = cv2.imread(fname)
    gray = cv2.cvtColor(img, cv2.COLOR_BGR2GRAY)
    ret, corners = cv2.findChessboardCorners(gray, (nx, ny), None)
    if ret:                                          # skip images where t
        corners = cv2.cornerSubPix(gray, corners, (11, 11), (-1, -1
        objpoints.append(objp)                       # same for every imag
        imgpoints.append(corners)

print(f"board found in {len(imgpoints)} / {len(images)} images")
```



All 1680 corners from the 35 GoPro images: they cover the **whole frame**, corners included. That is what makes a calibration reliable.

Calibration, step 2: `cv2.calibrateCamera`

▶ Lab 6C in Colab

```
cv2.calibrateCamera(objpoints,
imgpoints, gray.shape[::-1],None,None)
```

Return values:
ret, mtx, dist, rvecs, tvecs

```python
ret, mtx, dist, rvecs, tvecs = cv2.calibrateCamera(
    objpoints, imgpoints, gray.shape[::-1], None, None)

print("RMS reprojection error:", ret)  # 0.82 px
print(mtx)   # [[560.0   0.0  651.1]
             #  [  0.0 561.1  498.9]
             #  [  0.0   0.0    1.0]]
print(dist)  # [[-0.233  0.062  0.000  0.000 -0.008]]
```

Image size: (width, height)!

`gray.shape` is **(960, 1280)** = (rows, cols). OpenCV wants **(1280, 960)** = (w, h), hence `gray.shape[::-1]`.

For a colour image use `img.shape[1::-1]`. Passing the full `(960, 1280, 3)` raises an error.

What comes back

- `ret` : RMS reprojection error (px)
- `mtx` : K (3 × 3); `dist` : (1 × 5)
- `rvecs, tvecs` : one board pose per image

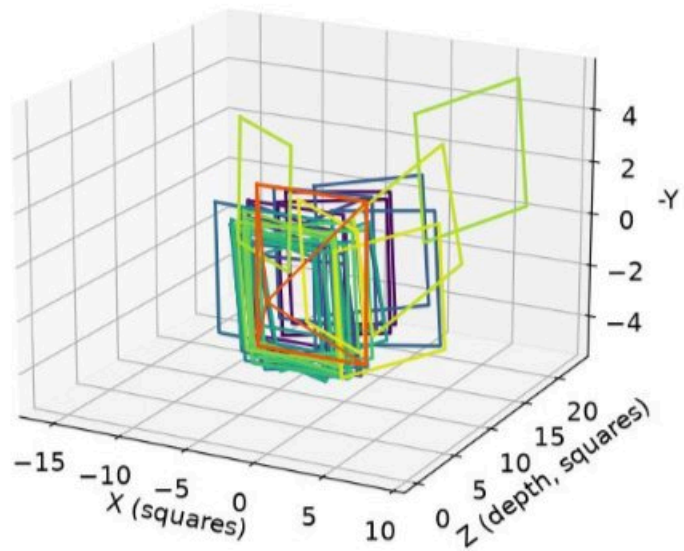
What happens inside: Zhang's method (2000)

▶ Lab 6M in Colab

1. For each view, the flat board (Z = 0) maps to the image by a **homography**: $\mathbf{x} \simeq H\,[X\ Y\ 1]^\top$, with $H = K\,[\mathbf{r}_1\ \mathbf{r}_2\ \mathbf{t}]$.
2. Columns r_1, r_2 of a rotation are **orthonormal**, which gives 2 equations per view on $B = K^{-\top}K^{-1}$:
   $\mathbf{h}_1^\top B\,\mathbf{h}_2 = 0, \quad \mathbf{h}_1^\top B\,\mathbf{h}_1 = \mathbf{h}_2^\top B\,\mathbf{h}_2$
3. B has 5 unknowns (zero skew: 4) → **≥ 3 views** (≥ 2 with zero skew) give a **closed-form K**; then each view's R, t.
4. Non-linear **Levenberg–Marquardt** refinement of K, dist and all poses together, minimising the **reprojection error**.

Only a printed board, no expensive 3-D rig, so it is used everywhere: OpenCV, MATLAB, ROS.

Extrinsics: where each board was, seen from the camera



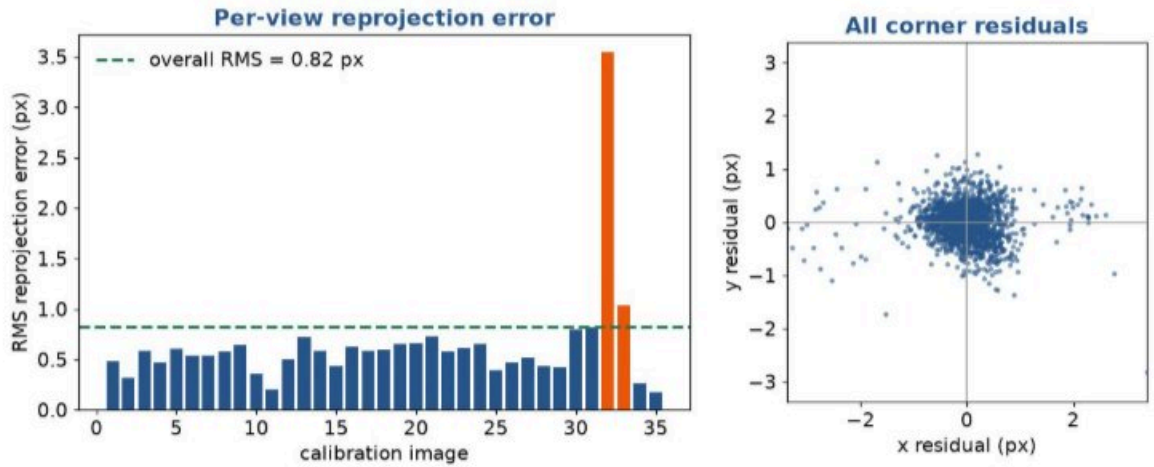
The 35 recovered board poses (rvecs, tvecs) seen from the camera (orange).

Watch: Camera Calibration using Zhang's Method and Intrinsic Camera Calibration (Cyrill Stachniss).

In plain words: every tilted photo of the flat board gives two clues about K. Three photos give enough clues to solve K; then a computer polishes all numbers together.

Is the calibration any good? Reprojection error

▶ Lab 6C in Colab



$$E_{RMS} = \sqrt{\frac{1}{N}\sum_i \|\mathbf{x}_i - \hat{\mathbf{x}}_i(K, \text{dist}, R_j, \mathbf{t}_j)\|^2}$$

Project the object points with the estimated model and compare with the detected corners.

- **< 0.5 px**: good; **0.5–1 px**: acceptable; **> 1 px**: investigate
- One view at **3.5 px** (orange): blurred or mis-detected. **Drop it and recalibrate** (Lab 6C).
- Low error ≠ correct: 3 nearly identical views can also fit well. Check **coverage**, too.

In plain words: put the board back into the picture using the model and measure how far the predicted corners are from the real ones. Small = the model explains the photos well.

Per-view error and saving the result

▶ Lab 6C in Colab

```
errs = []
for objp_i, imgp_i, r, t in zip(objpoints, imgpoints,
                                rvecs, tvecs):
    proj, _ = cv2.projectPoints(objp_i, r, t, mtx, dist)
    d = imgp_i.reshape(-1, 2) - proj.reshape(-1, 2)
    errs.append(np.sqrt((d ** 2).sum(1).mean()))

worst = np.argsort(errs)[::-1][:3]
print("worst views:", [images[i] for i in worst])

# Uncertainty of each intrinsic (std. dev.)
ret, mtx, dist, rv, tv, std_in, std_ex, per_view = \
    cv2.calibrateCameraExtended(objpoints, imgpoints,
                                size, None, None)
print("fx = %.1f ± %.1f px" % (mtx[0, 0], std_in[0, 0]))
```

```
# Save: portable YAML that C++, ROS and Python can read
fs = cv2.FileStorage("camera.yaml", cv2.FILE_STORAGE_WRITE
fs.write("image_width", size[0]); fs.write("image_height",
fs.write("camera_matrix", mtx)
fs.write("distortion_coefficients", dist)
fs.write("rms", ret)
fs.release()

# Load it back
fs = cv2.FileStorage("camera.yaml", cv2.FILE_STORAGE_READ)
K = fs.getNode("camera_matrix").mat()
D = fs.getNode("distortion_coefficients").mat()

# or simply
np.savez("camera.npz", K=mtx, dist=dist)
# (Udacity used pickle: wide_dist_pickle.p)
```

How to take good calibration images

Coverage & variety

- **15–30 images** (Udacity: ≥ 20)
- Board in every **part of the frame**, especially the **corners**, where distortion is largest
- **Tilt** the board ±30–45° in both axes. Frontal-only views cannot separate f from distance
- Vary the **distance**: near (board fills the frame) and far

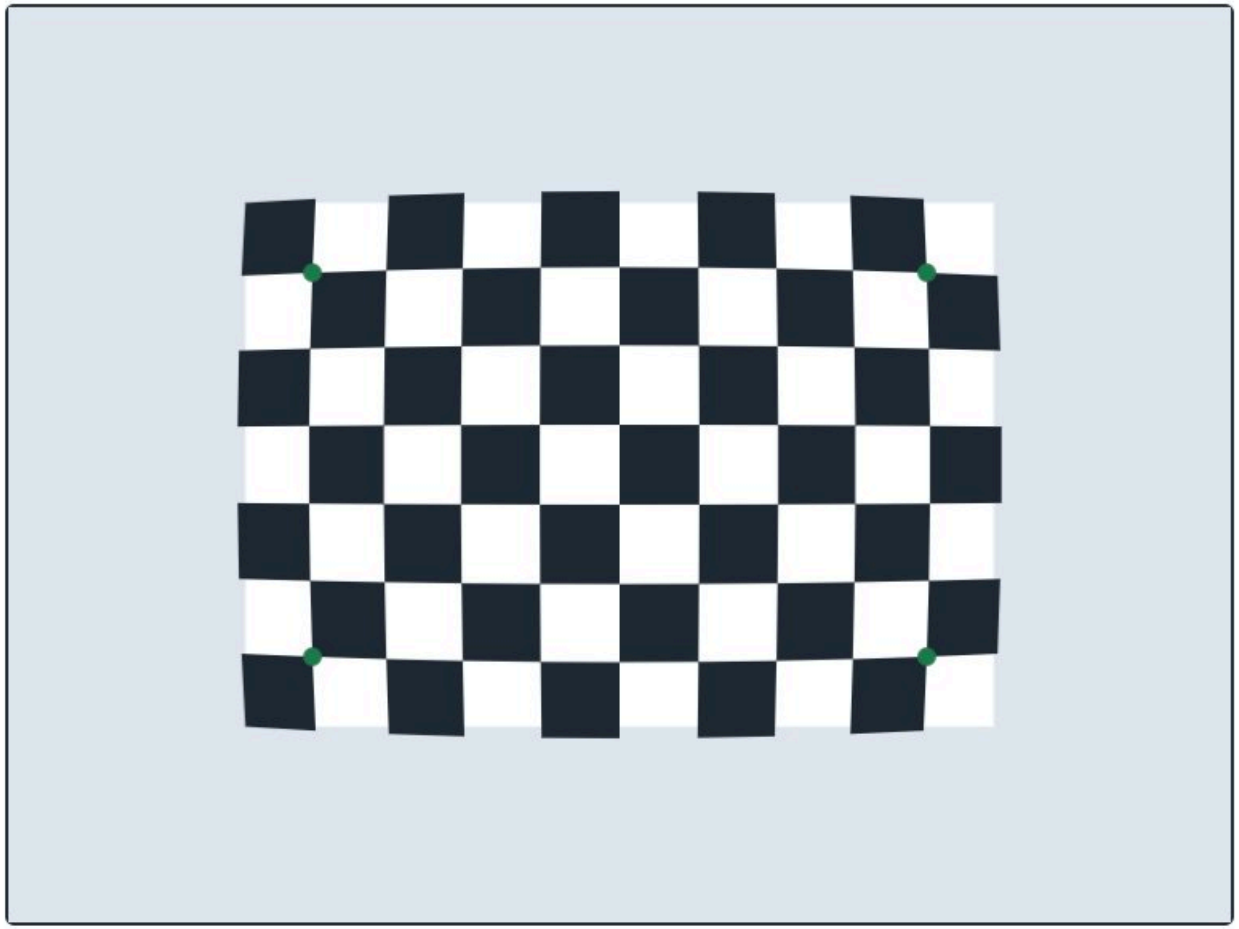
Quality

- Sharp: **still** board, good light, short exposure
- Same **resolution, focus and zoom** you will use later: K changes with all three
- Do not crop or resize the images before calibration
- Check the **per-view error** and coverage, re-shoot the gaps

The **ROS** calibrator turns this advice into four progress bars, **X, Y, Size and Skew**, and only enables CALIBRATE once your views cover enough of each. Try it on the next slide.

Live demo: collect a good calibration set interactive

▶ Lab 6I in Colab



board x 0

board y 0

distance 1.00

tilt left/right 0

tilt up/down 0

📷 capture | random pose | clear

X 0%

Y 0%

Size 0%

Skew 0%

0 samples · CALIBRATE (not yet)

Bars follow ROS `camera_calibration`: the spread of the board's centre (X, Y), apparent size and skew over the accepted samples. Samples too similar to an earlier one are rejected.

The same idea in ROS 2: camera_calibration

```
# install (ROS 2 Humble / Jazzy / Rolling)
sudo apt install ros-$ROS_DISTRO-camera-calibration

# monocular: 8x6 inner corners, 25 mm squares
ros2 run camera_calibration cameracalibrator \
  --size 8x6 --square 0.025 \
  --ros-args -r image:=/camera/image_raw \
             -p camera:=/camera

# stereo pair
ros2 run camera_calibration cameracalibrator \
  --size 8x6 --square 0.025 --approximate 0.1 \
  --ros-args -r left:=/left/image_raw -r right:=/right/image_raw

# fisheye lens:   --camera-model fisheye
# ChArUco board:  --pattern charuco --charuco_marker_size 0.019 --a
```

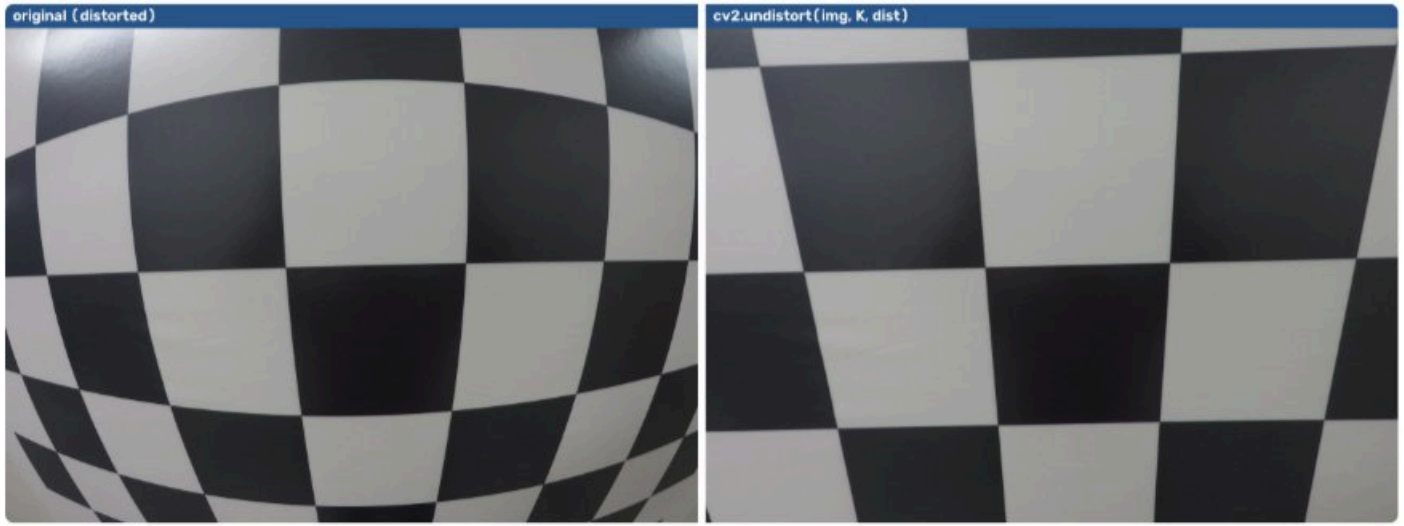
How it works

- Uses OpenCV under the hood: the Python module `camera_calibration.calibrator` (`MonoCalibrator` , `StereoCalibrator`)
- Move the board until the X / Y / Size / Skew bars go **green** → **CALIBRATE** → **SAVE** / **COMMIT**
- Writes a `camera_info` **YAML** (K, D, R, P), loaded by `camera_info_manager` and used by `image_proc` to publish `image_rect`
- Models: `pinhole` (plumb-bob) and `fisheye` (equidistant)

docs.ros.org: camera_calibration

Calibration, step 3: undistort

▶ Lab 6C in Colab




```
dst = cv2.undistort(img, mtx,
dist, None, mtx)
```

- Takes a distorted image, K and dist; returns the corrected **destination** image
- The last argument is the **new** camera matrix; passing `mtx` keeps the same focal length
- For every output pixel OpenCV computes where it came from (the distortion model) and interpolates

> Straight lines on the board are straight again: the calibration worked.

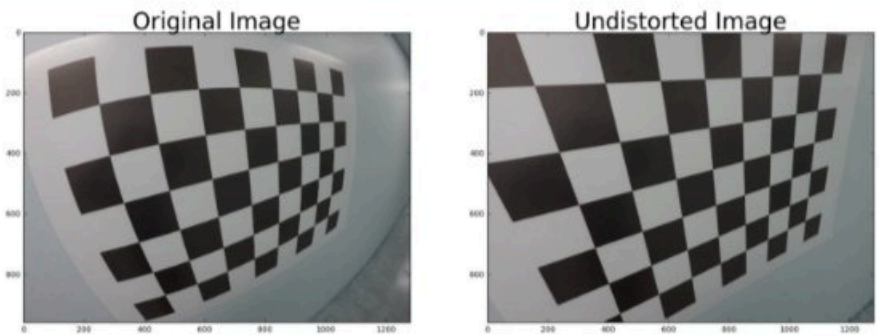
Quiz: write `cal_undistort()`

▶ Lab 6C in Colab

```python
import pickle, cv2, numpy as np

# Saved objpoints and imgpoints (Udacity wide_dist_pickle.
dist_pickle = pickle.load(open("wide_dist_pickle.p", "rb"))
objpoints = dist_pickle["objpoints"]
imgpoints = dist_pickle["imgpoints"]
img = cv2.imread("test_image.png")

# TODO: take an image, object points and image points,
# calibrate, undistort and return the undistorted image
def cal_undistort(img, objpoints, imgpoints):
    undist = np.copy(img)        # delete this line
    return undist
```

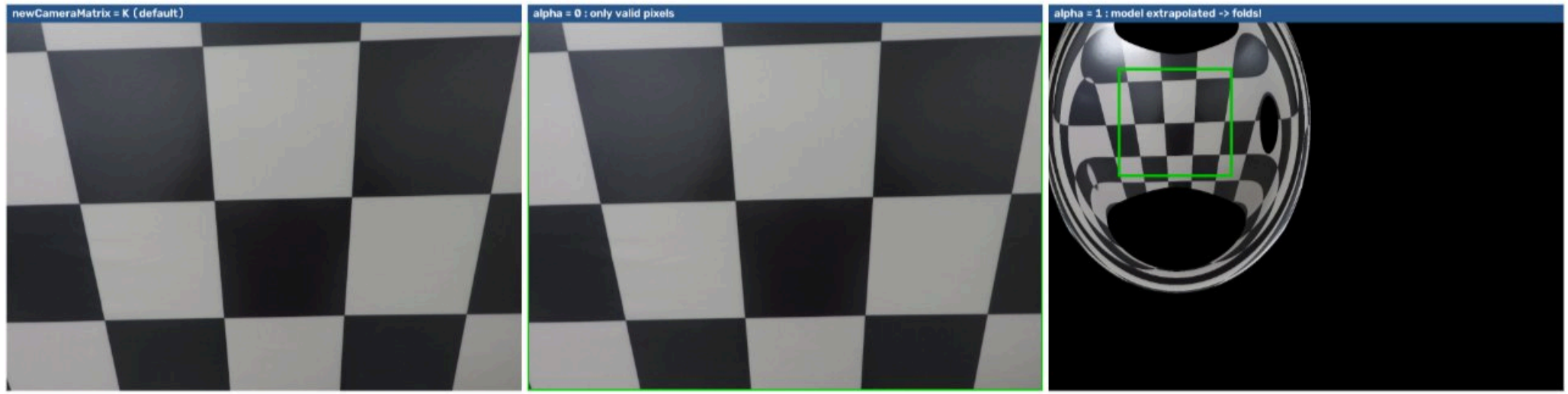


One solution

```python
def cal_undistort(img, objpoints, imgpoints):
    gray = cv2.cvtColor(img, cv2.COLOR_BGR2GRAY)
    ret, mtx, dist, rvecs, tvecs = cv2.calibrateCamera(
        objpoints, imgpoints, gray.shape[::-1], None, None)
    return cv2.undistort(img, mtx, dist, None, mtx)
```

In practice calibrate **once**, save mtx and dist, and only call `undistort` per frame. Lab 6C has an auto-check.

Keep or crop the borders? `getOptimalNewCameraMatrix`




```
h, w = img.shape[:2]
newK, roi = cv2.getOptimalNewCameraMatrix(
    mtx, dist, (w, h), alpha=0)   # 0 … 1
und = cv2.undistort(img, mtx, dist, None, newK)
x, y, rw, rh = roi                  # green box
und = und[y:y + rh, x:x + rw]       # crop black edges
```

- **alpha = 0**: zoom in until **every** output pixel is valid (loses a little FOV)
- **alpha = 1**: keep **all** source pixels (black curved borders)
- Here alpha = 1 **folds**: the 5-coefficient polynomial is only valid where there were corners. Try `CALIB_RATIONAL_MODEL` or fisheye for wide lenses (Lab 6C).

▶ Lab 6C in Colab

Undistorting video fast: compute the map once

▶ Lab 6C in Colab

```python
# once, at start-up
newK, roi = cv2.getOptimalNewCameraMatrix(K, dist, (w, h), 0)
map1, map2 = cv2.initUndistortRectifyMap(
    K, dist, None, newK, (w, h), cv2.CV_16SC2)

cap = cv2.VideoCapture(0)          # or Picamera2 frames
while True:
    ok, frame = cap.read()
    if not ok: break
    und = cv2.remap(frame, map1, map2, cv2.INTER_LINEAR)
    cv2.imshow("undistorted", und)
    if cv2.waitKey(1) == 27: break

# undistort only some points (e.g. detected corners, lane pixels)
pts_ud = cv2.undistortPoints(pts, K, dist, P=newK)
```

Why it is faster

`cv2.undistort` recomputes the distortion model for every pixel on every call.

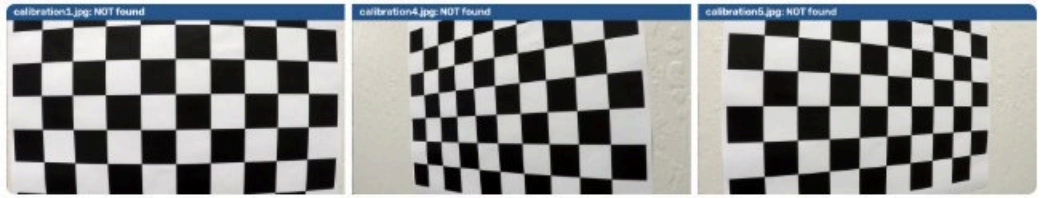
`initUndistortRectifyMap` stores the source coordinate of each output pixel **once**; `remap` is then a pure lookup: several times faster, which matters at 30 fps on a Raspberry Pi.

If you only need a few points, `undistortPoints` avoids warping the image at all.

Common calibration pitfalls

▶ Lab 6F in Colab

Symptom	Likely cause
`ret` is `False` for many images	wrong (nx, ny): squares instead of inner corners; board cut off by the frame; glare
`calibrateCamera` error about sizes	passed `img.shape` with 3 channels, or images of **different sizes** (two Udacity images are 1281 × 721!)
RMS > 1 px	blur, a bent board, a mis-detected view: check per-view errors
Edges still curved / folded after undistort	no corners near the image edges; lens too wide for the model → k3, rational, or fisheye
Good RMS, wrong distances	wrong `square_size`; calibrated at another resolution or focus
Image upside-down / mirrored axes	symmetric board (8 × 8): use nx ≠ ny



3 of the 20 road-camera images fail: part of the board is **outside the frame**. Fine for a chessboard: they are skipped. A ChArUco board would still use them.

Lab 6F is a bug hunt with these pitfalls planted in the code.

REMEMBER

Calibrating with OpenCV: the 3 things to remember

1. Corners: `findChessboardCorners` + `cornerSubPix`, count the **inner** corners.
2. `calibrateCamera(objpoints, imgpoints, (w, h))` → K, dist and one pose per photo.
3. Judge it with the **RMS reprojection error** (< 0.5 px good, > 1 px re-shoot), then `undistort` / `remap`.

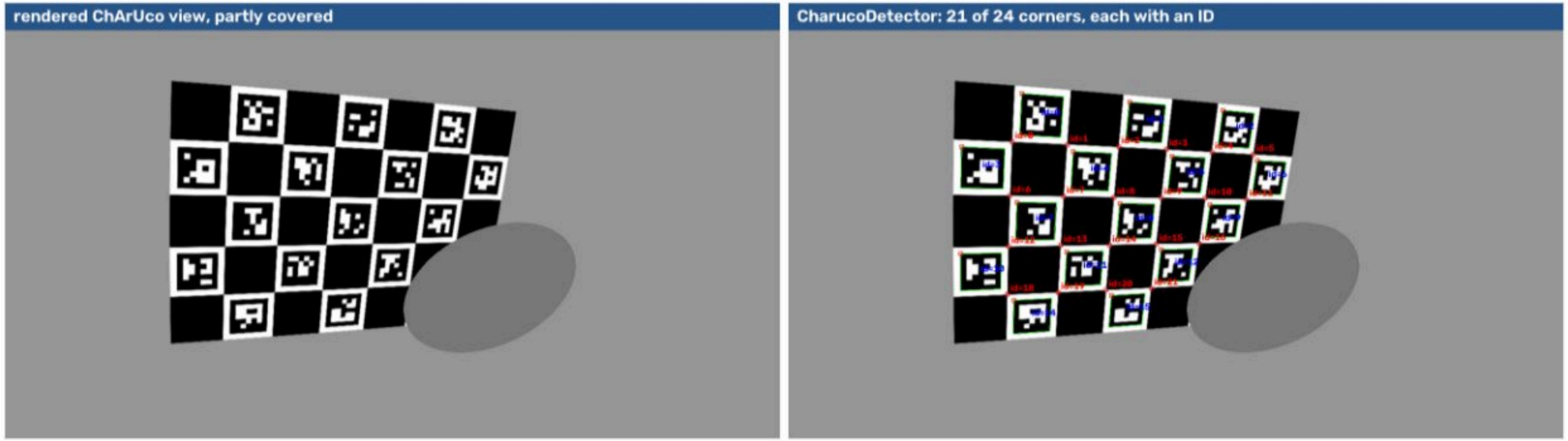
06

Beyond the basic chessboard

ChArUco boards · fisheye lenses · stereo pairs · your own Raspberry Pi camera

ChArUco: a chessboard with an ID on every square

▶ Lab 6E in Colab



- ArUco markers inside the white squares tell the detector **which** corner it is seeing
- So **partial, occluded or out-of-frame** boards still give corners: perfect for covering image **edges** and fisheye lenses
- Corners are chessboard **saddle points** → as accurate as a chessboard

Watch: How to do Camera Calibration with ChArUco Boards (Nicolai Nielsen) · OpenCV tutorial: calibration with ArUco and ChArUco

ChArUco calibration: the code (OpenCV ≥ 4.7 / 5)

▶ Lab 6E in Colab

```python
import cv2, glob

aruco_dict = cv2.aruco.getPredefinedDictionary(cv2.aruco.DICT_5X5_1
board = cv2.aruco.CharucoBoard((7, 5),          # squares x, y
                               0.040, 0.030,    # square, marker (m
                               aruco_dict)
cv2.imwrite("board.png", board.generateImage((2100, 1500), marginSi

detector = cv2.aruco.CharucoDetector(board)
all_obj, all_img = [], []
for f in glob.glob("charuco/*.jpg"):
    gray = cv2.imread(f, cv2.IMREAD_GRAYSCALE)
    ch_corners, ch_ids, m_corners, m_ids = detector.detectBoard(gra
    if ch_ids is not None and len(ch_ids) >= 6:     # partial view
        obj, img = board.matchImagePoints(ch_corners, ch_ids)
        all_obj.append(obj); all_img.append(img)

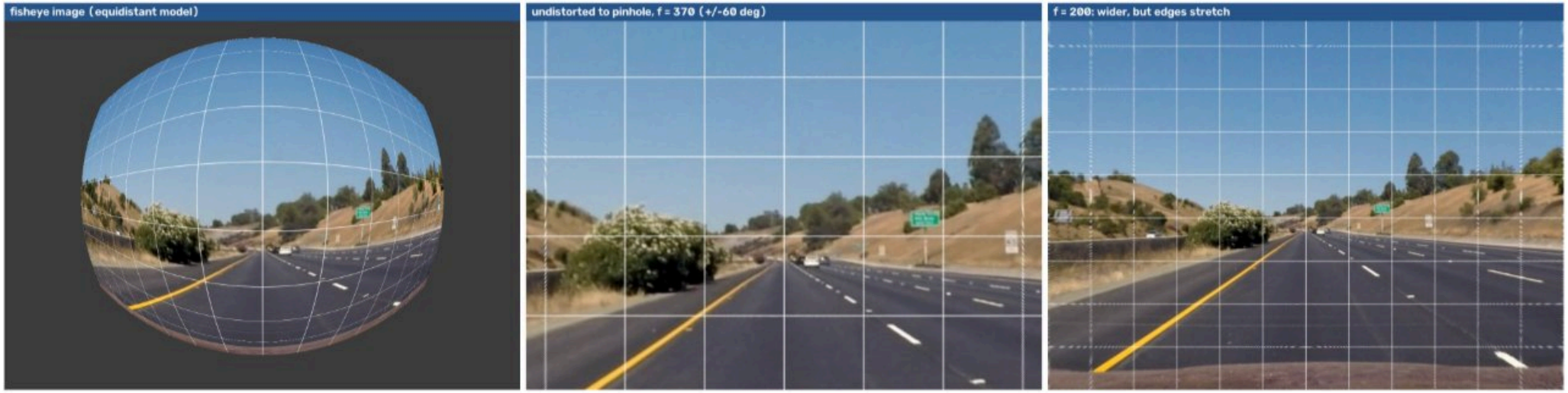
rms, K, dist, rvecs, tvecs = cv2.calibrateCamera(
    all_obj, all_img, gray.shape[::-1], None, None)
```

Notes

- The old `cv2.aruco.calibrateCameraCharuco` / `interpolateCornersCharuco` were removed in 4.7+. Use `CharucoDetector` + `matchImagePoints`.
- Measure both the square and marker size on the print.
- Pick a dictionary with few markers (e.g. 5×5_100) for better robustness.
- Lab 6E renders synthetic ChArUco views with a **known** camera, so you can check your calibration against the ground truth.

Fisheye lenses: cv2.fisheye

▶ Lab 6E in Colab



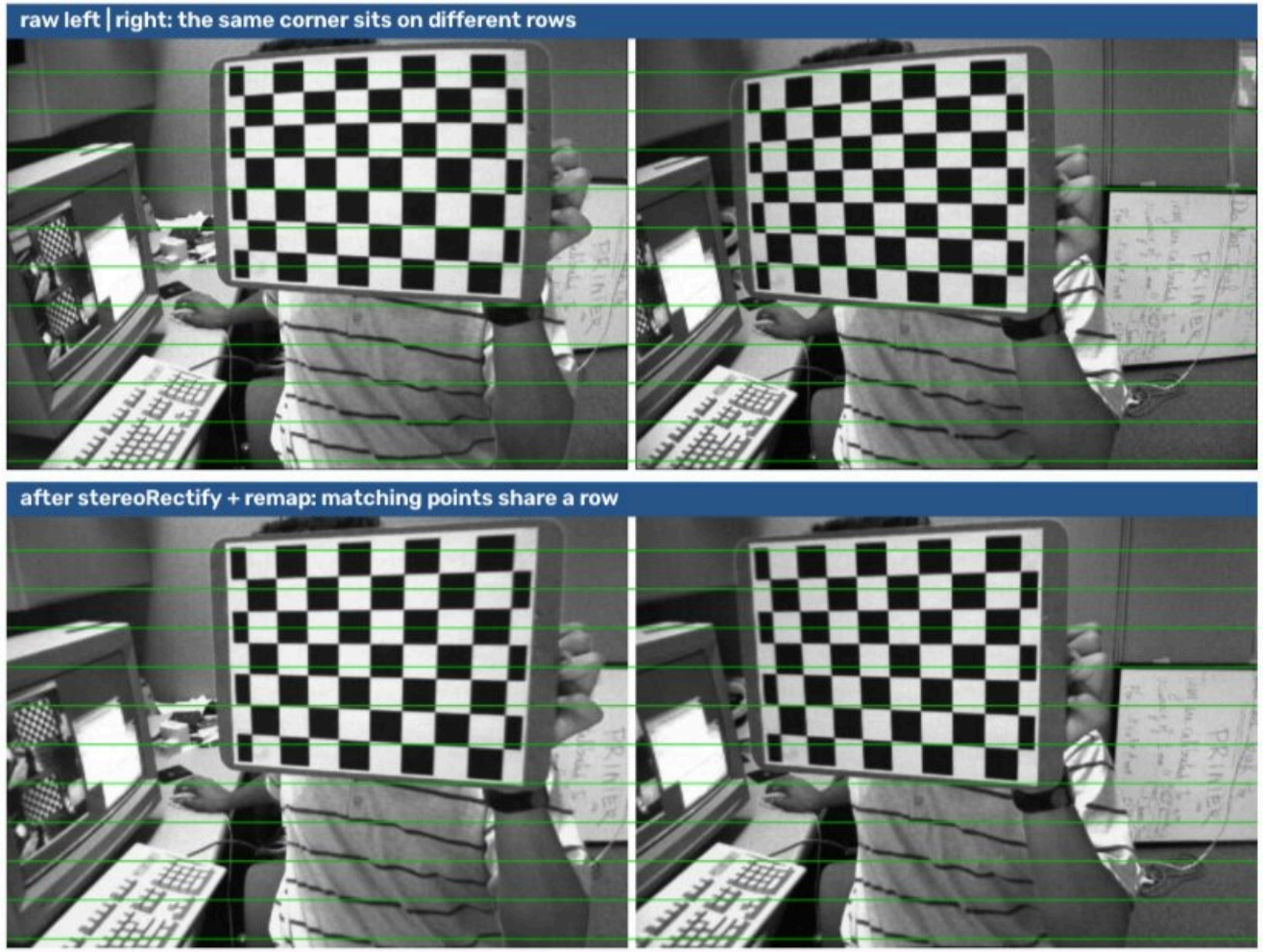

```
# per view: objpoints (1,N,3) float64, imgpoints (1,N,2) float64
# OpenCV 5: cv2.CALIB_*   (OpenCV 4: cv2.fisheye.CALIB_*)
flags = cv2.CALIB_RECOMPUTE_EXTRINSIC | cv2.CALIB_FIX_SKEW
rms, K, D, rvecs, tvecs = cv2.fisheye.calibrate(
    objpoints, imgpoints, (w, h), None, None, flags=flags)   # D =

newK = cv2.fisheye.estimateNewCameraMatrixForUndistortRectify(
    K, D, (w, h), np.eye(3), balance=0.0)                    # 0 cr
m1, m2 = cv2.fisheye.initUndistortRectifyMap(K, D, np.eye(3), newK,
und = cv2.remap(img, m1, m2, cv2.INTER_LINEAR)
```

- The whole 180° view **cannot** fit into one pinhole image: choose a narrower **newK** (middle) or accept stretched edges (right)
- Points must be float64, shape **(1, N, 3)**: OpenCV 5 rejects the old (N, 1, 3). The #1 fisheye error
- Very wide (> 200°) or mirror cameras: `cv2.omnidir` (contrib)

Stereo calibration: two cameras, one rig

▶ Lab 6G in Colab



1. Capture the board with **both cameras at the same moment** (hardware sync or a still board)
2. Calibrate each camera → K_1, D_1, K_2, D_2
3. `stereoCalibrate` → **R, T** between the cameras (+ E, F)
4. `stereoRectify` → rotations that make epipolar lines **horizontal**
5. `initUndistortRectifyMap` + `remap` per frame → disparity → depth

OpenCV's left/right sample pair: RMS 0.45 px, baseline **3.3 squares** along −x.

Depth from disparity, $Z = fB/d$, is Lecture 10. Datasets: Kaggle stereo chessboard pictures.

Stereo calibration: the code

▶ Lab 6G in Colab

```
# objpoints, imgL, imgR: corners found in BOTH images of each pair
_, K1, D1, _, _ = cv2.calibrateCamera(objpoints, imgL, size, None, None)
_, K2, D2, _, _ = cv2.calibrateCamera(objpoints, imgR, size, None, None)

rms, K1, D1, K2, D2, R, T, E, F = cv2.stereoCalibrate(
    objpoints, imgL, imgR, K1, D1, K2, D2, size,
    flags=cv2.CALIB_FIX_INTRINSIC)           # keep the mono results, solve only R, T
print("stereo RMS", rms, "baseline", np.linalg.norm(T) * square_mm, "mm")

R1, R2, P1, P2, Q, roi1, roi2 = cv2.stereoRectify(K1, D1, K2, D2, size, R, T, alpha=0)
mapL = cv2.initUndistortRectifyMap(K1, D1, R1, P1, size, cv2.CV_16SC2)
mapR = cv2.initUndistortRectifyMap(K2, D2, R2, P2, size, cv2.CV_16SC2)
rectL = cv2.remap(left,  *mapL, cv2.INTER_LINEAR)
rectR = cv2.remap(right, *mapR, cv2.INTER_LINEAR)       # now matching points share a row
```

Two Raspberry Pi cameras on a bracket (or the Pi 5's two CSI ports) make a cheap stereo rig. Keep them **rigidly** mounted: any movement invalidates R, T. See Raspberry Pi stereo camera calibration.

Check the rectified pair: draw horizontal lines. If the same corner is on different rows, the calibration or the synchronisation is off.

Challenge: calibrate your own camera

▶ Lab 6H in Colab

```python
# capture_calib.py  (Lab 6H): Pi camera or USB webcam
from picamera2 import Picamera2            # USB: cv2.VideoCapture(0
import cv2, time, os

picam2 = Picamera2()
picam2.configure(picam2.create_still_configuration(
    main={"size": (1920, 1080), "format": "RGB888"}))
picam2.start()
picam2.set_controls({"AfMode": 0, "LensPosition": 0.0})  # Module 3

os.makedirs("calib_images", exist_ok=True)
for i in range(25):
    time.sleep(3)                          # move the board, hold
    frame = picam2.capture_array()
    gray = cv2.cvtColor(frame, cv2.COLOR_BGR2GRAY)
    ok, _ = cv2.findChessboardCorners(gray, (9, 6),
                cv2.CALIB_CB_FAST_CHECK)
    cv2.imwrite(f"calib_images/img_{i:02d}.jpg", frame)
    print(i, "board found" if ok else "NO board: move it")
```

Recipe

1. Print a 9 × 6 board (pattern.png), stick it to a flat board, measure a square
2. Take 20–30 pictures over the **whole field of view**, especially near the edges
3. Run `calibrate.py --board 9x6 --square 24.5` → `camera.yaml`
4. Check RMS < 0.5–1 px and that straight lines are straight
5. Commit `camera.yaml` next to your project code

Based on Petros626/Camera-Calibration-with-PiCamera2-OpenCV, niconielsen32/camera-calibration, nuwandda/Checkerboard-Calibration, ibois-epfl/rgb-camera-calibration.

Calibration tools in the wild

Tool	Best for	Notes
OpenCV calib3d (`cv2.calibrateCamera` , `fisheye` , `stereoCalibrate` , `aruco`)	everything in this course	calib3d reference: models, flags, functions
OpenCV `interactive-calibration` app	guided live calibration	suggests the next pose, shows per-parameter uncertainty
ROS 2 `camera_calibration`	robots, mono/stereo, fisheye, ChArUco	X/Y/Size/Skew bars → `camera_info` YAML
MATLAB Camera Calibrator / Stereo Camera Calibrator	teaching, quick visual checks	error plots and 3-D pose views
Kalibr (ETH Zürich)	multi-camera + IMU, rolling shutter	AprilGrid targets; the standard for VIO / SLAM rigs
calib.io camera calibrator, **mrcal** (NASA JPL)	high-accuracy industrial / metrology	richer lens models, uncertainty maps

Whatever the tool, the inputs and outputs are the same: **known 3-D target points + detected 2-D corners → K, distortion, poses, reprojection error.**

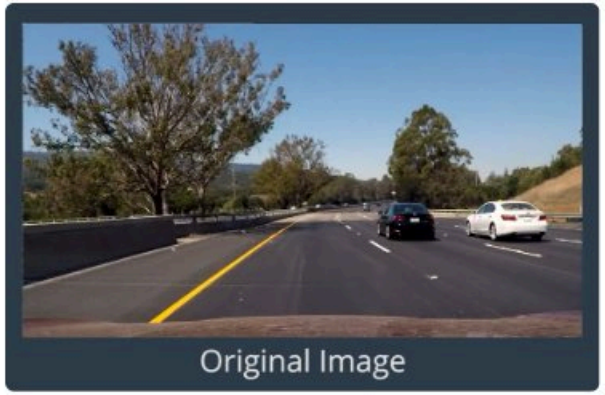
07

Perspective transform & lane curvature

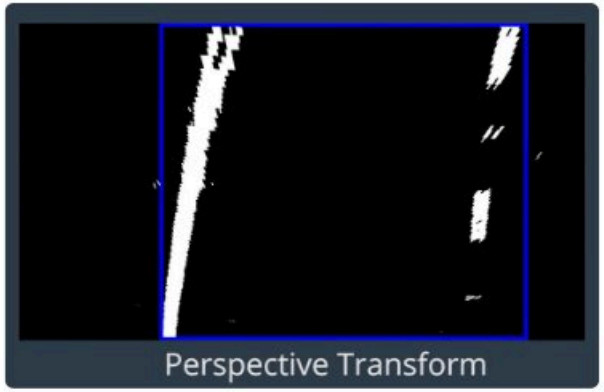
Using the undistorted image to measure the road

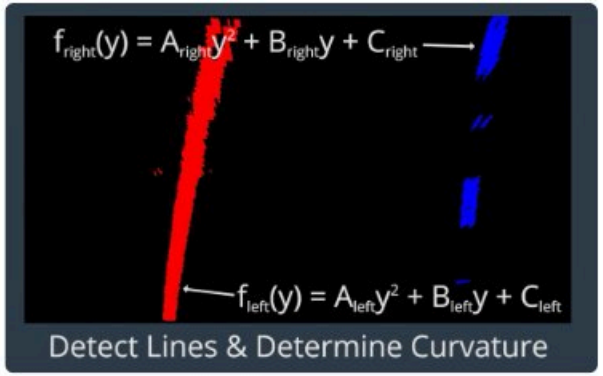
Why we need all this: lane curvature

▶ Lab 6D in Colab


Original Image


Thresholded Image


Perspective Transform



Detect Lines & Determine Curvature

A self-driving car needs a **steering angle**, which comes from the car's speed and dynamics and **how much the lane curves:**

1. threshold the lane pixels (Lecture 5)
2. **perspective transform** → bird's-eye view
3. fit a polynomial to each line
4. compute the curvature

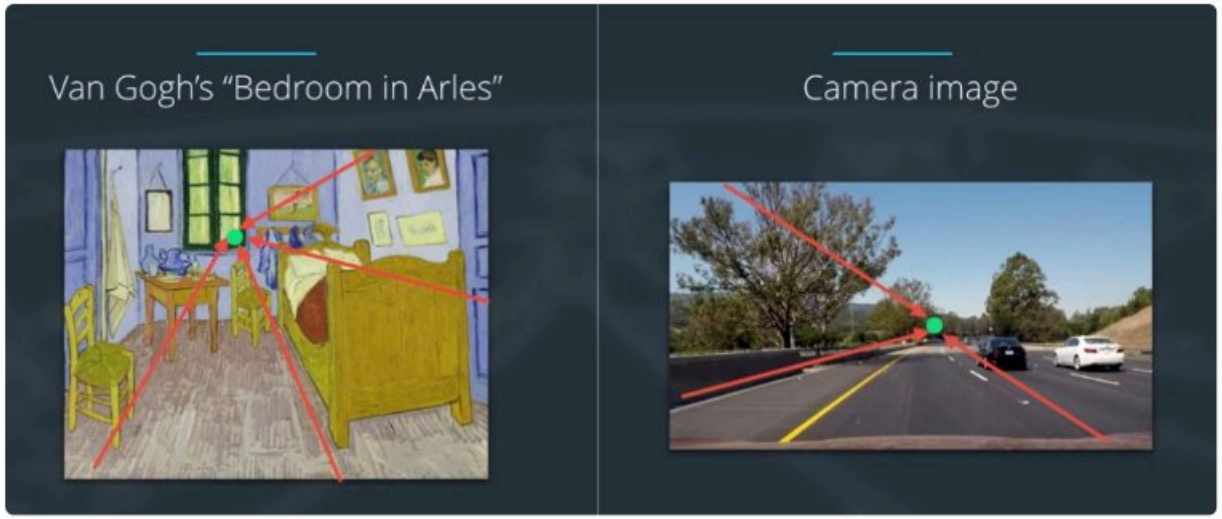
For a near-vertical lane line

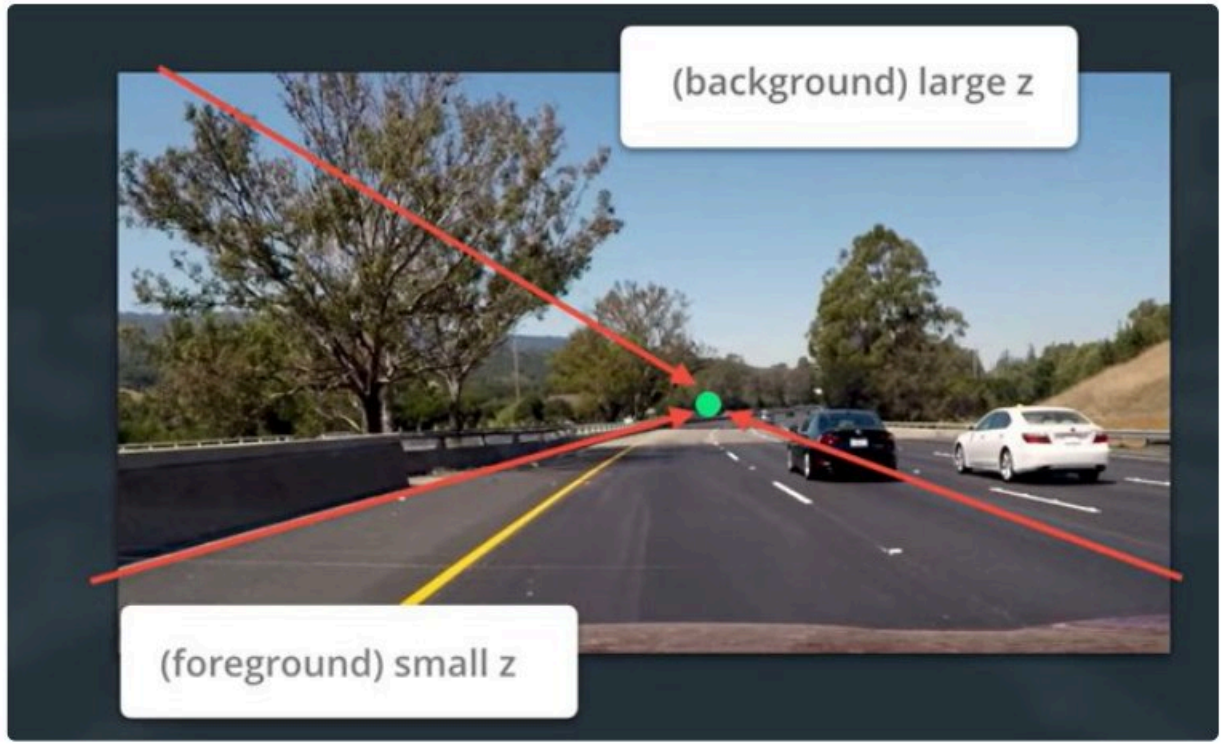
$$f(y) = Ay^2 + By + C$$

A → curvature · **B** → heading (direction) · **C** → position (offset from the left of the image)

$$R_{curve} = \frac{(1 + (2Ay + B)^2)^{3/2}}{|2A|}$$

Perspective: far things look small, parallel lines meet





The greater an object's **Z** (distance from the camera), the smaller it appears: the 1/Z of the pinhole model. Artists use the same rule. Lane lines that are **parallel** on the road **converge** to a vanishing point in the image.

A perspective transform changes the viewpoint



4 points on a plane in the image …

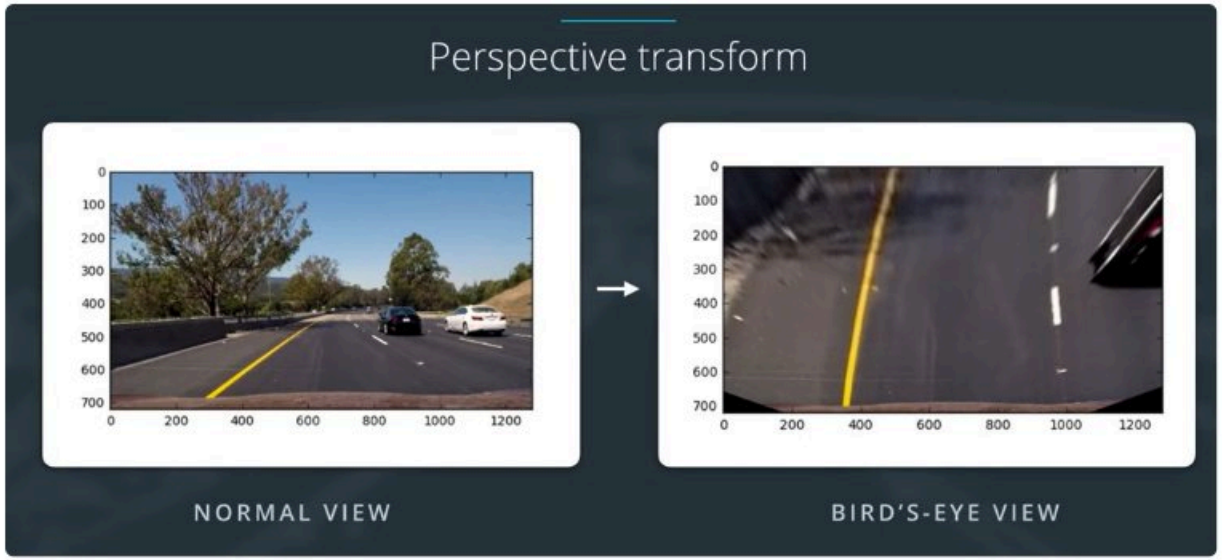


… mapped to 4 points of a rectangle

A perspective transform maps the points of a given image to **different, desired** image points, as if seen from a new viewpoint. It effectively drags points towards or away from the camera. **Four point pairs** are enough to define it (a 3 × 3 **homography** with 8 degrees of freedom).

> **In plain words:** four corners of a flat thing are enough to tell OpenCV how to "walk around" it and photograph it from the front or from above.

The bird's-eye view of a road





- From the car's camera, the left line seems to lean and curve to the right, the right line to lean left without much curve.
- From above, both lines are **parallel**, a **consistent width** apart, and curve **by the same amount**.
- A top-down view also lines up directly with a **map**.

Quiz: why a bird's-eye view?

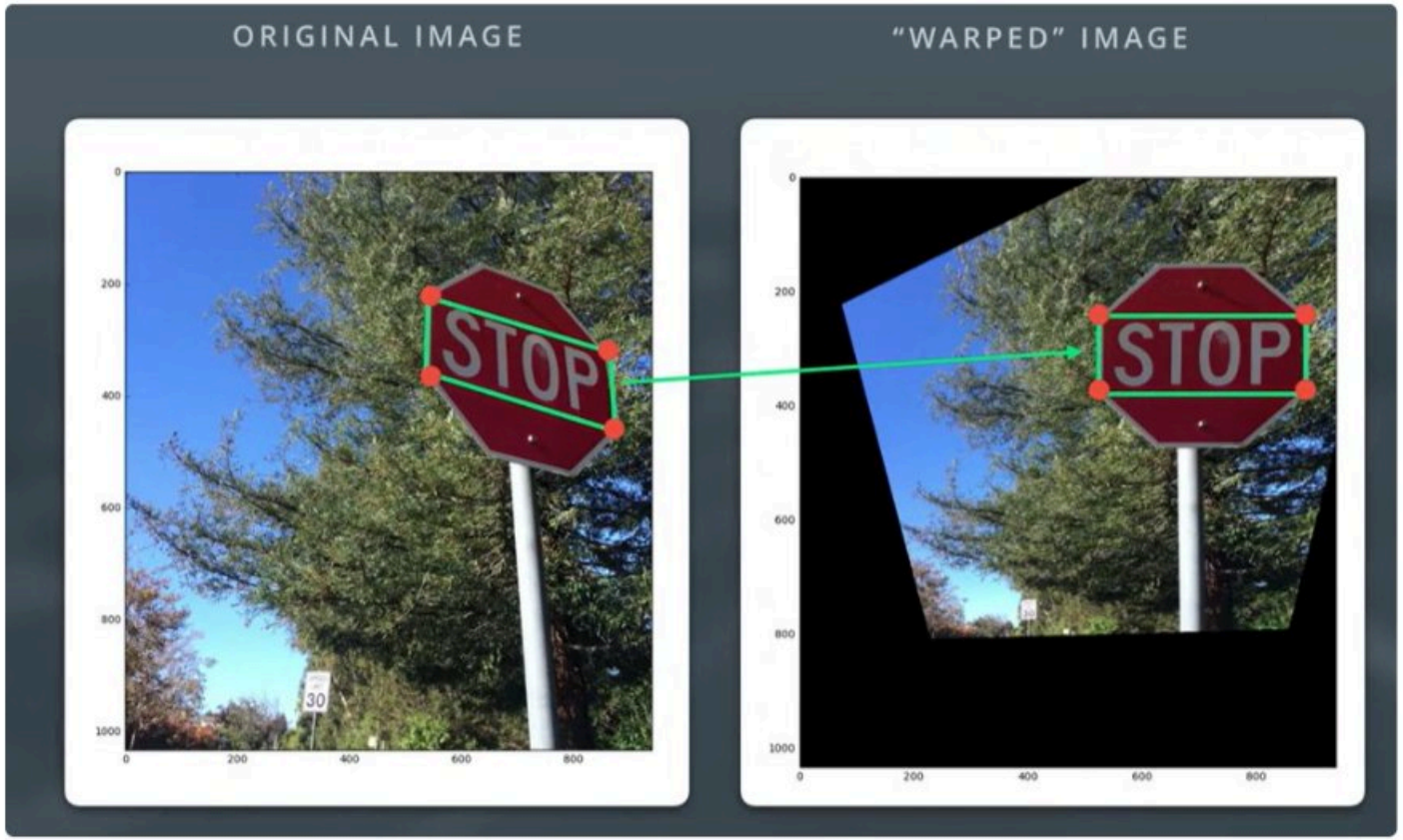
QUIZ QUESTION

For images from a front-facing car camera, why do we want to perform a perspective transform?

a. Because it is more straightforward to take a gradient on a top-down view.
b. Because ultimately we want to measure the **curvature** of the lines, and to do that we need to transform to a top-down view.
c. Because transforming the perspective removes the effect of curvature from our measurement.
d. Because it removes the effect of shadows and colour differences.

> **B.** Curvature is a property of the road seen from above. In the camera view perspective bends and squeezes it.

Example: transform a stop sign



1. Pick **4 source points** that define a rectangle on a **plane** in the image (the face of the sign)
2. Pick **4 destination points**: where they should appear in the **warped** image
3. `getPerspectiveTransform(src, dst)` → **M**
4. `warpPerspective(img, M, size)` → the warped image

A traffic sign is a good test because **text** shows clearly whether the transform is right.

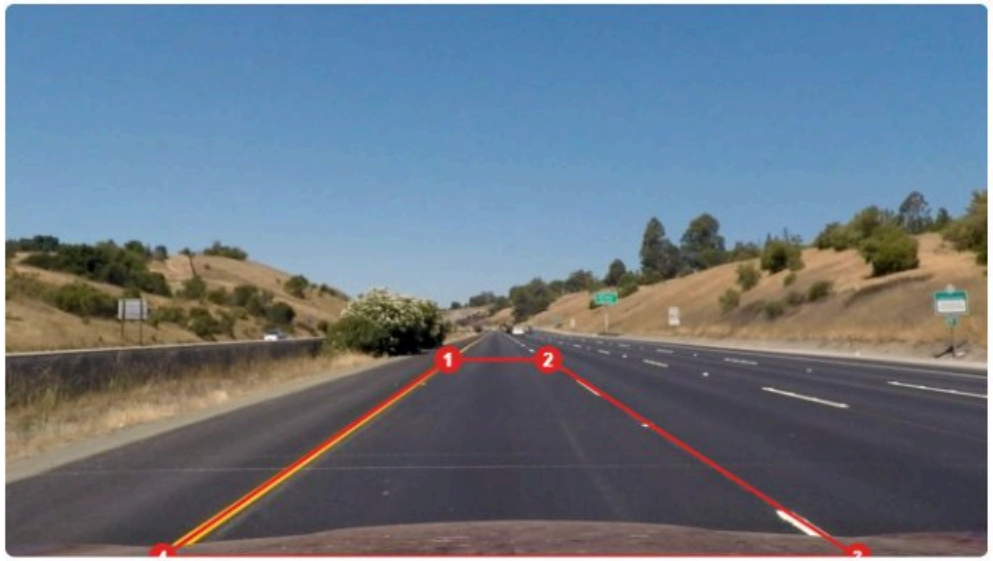
Perspective transform: the code

```python
def warp(img):
    img_size = (img.shape[1], img.shape[0])        # (w, h)
    # four source points on the sign (read off by hovering
    # the mouse over the image), clockwise from top-right
    src = np.float32([[850, 320], [865, 450],
                      [533, 350], [535, 210]])
    # where we want them: a front-facing rectangle
    dst = np.float32([[870, 240], [870, 370],
                      [520, 370], [520, 240]])
    M    = cv2.getPerspectiveTransform(src, dst)
    Minv = cv2.getPerspectiveTransform(dst, src)   # to un-warp
    warped = cv2.warpPerspective(img, M, img_size,
                                 flags=cv2.INTER_LINEAR)
    return warped
```

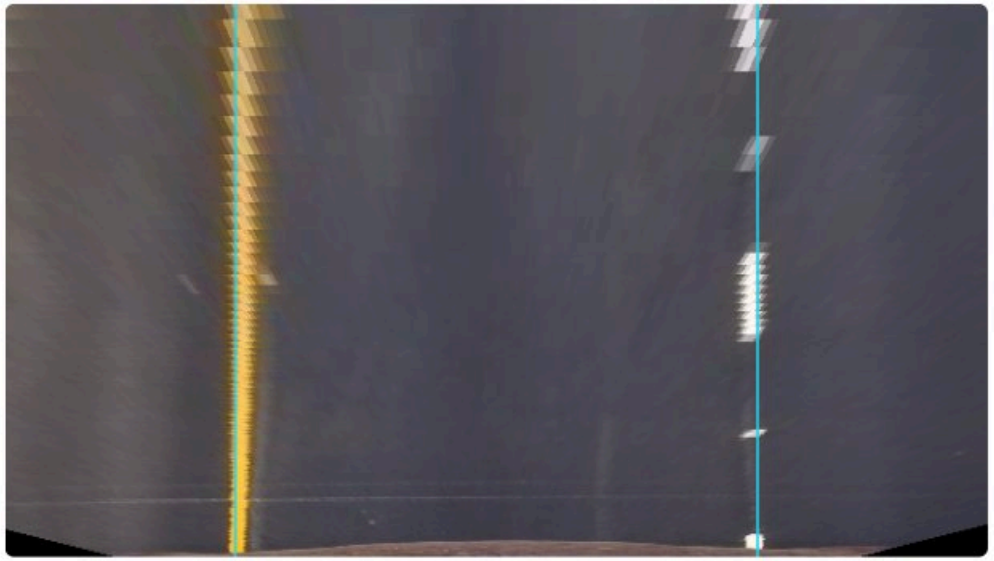


Choosing points **by hand** is fine for a fixed car camera (do it once). Many systems pick them **automatically** from corners, edges or detected objects, as you will with chessboard corners in the next quiz.

Live demo: drag four points → bird's-eye view interactive



Drag the 4 red handles onto the lane lines (undistorted frame, Lab 6D camera).



Warped: good points make the lane lines **vertical and parallel**.

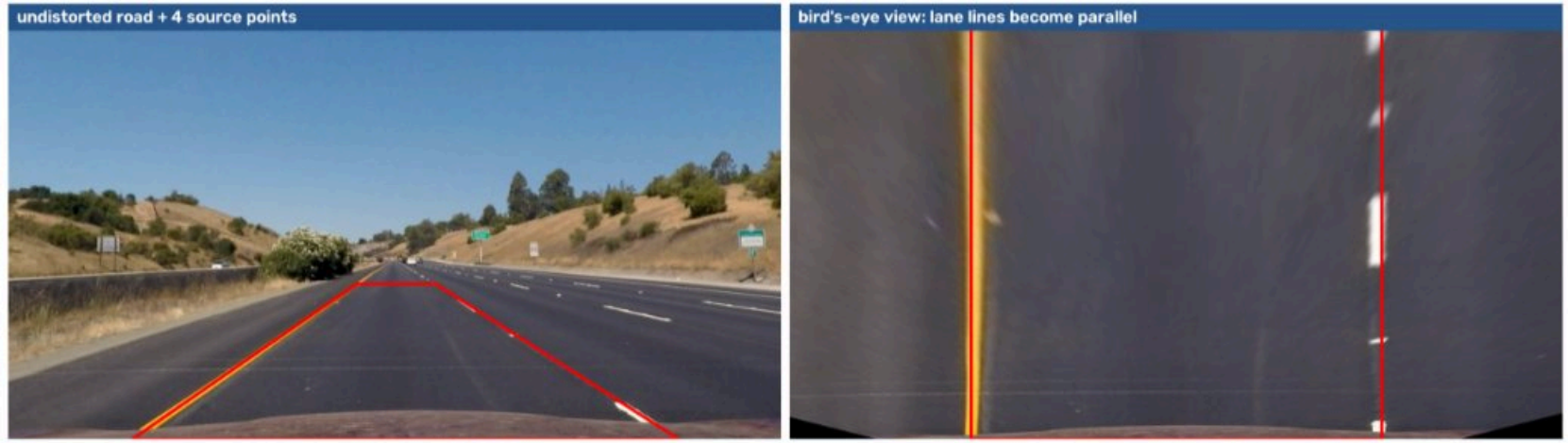
recommended points | a bad choice | toggle grid

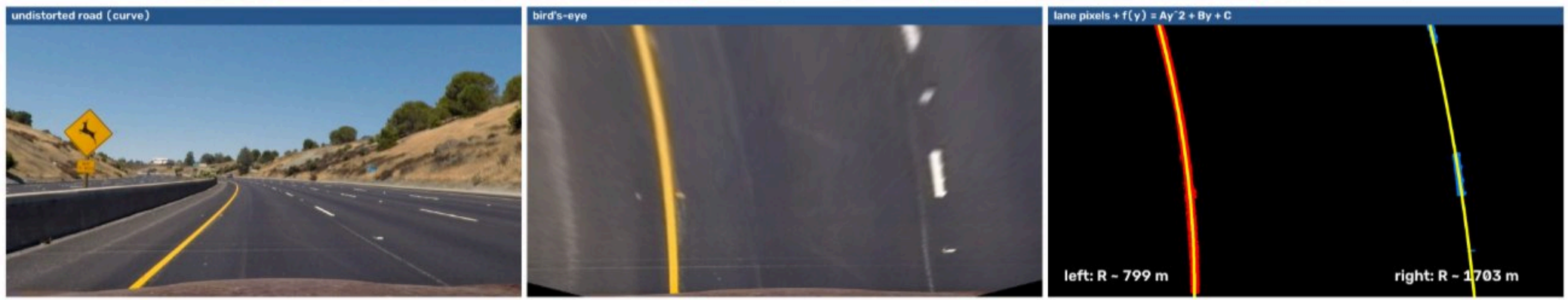
```
src = [[577,463],[707,463],[1110,720],[205,720]]
M = [[   -0.537,   -1.492,  969.782],
     [   -0.000,   -2.002,  927.096],
     [   -0.000,   -0.002,    1.000]]
```

▶ Lab 6D in Colab

Undistort → warp → fit: on the real road camera

▶ Lab 6D in Colab

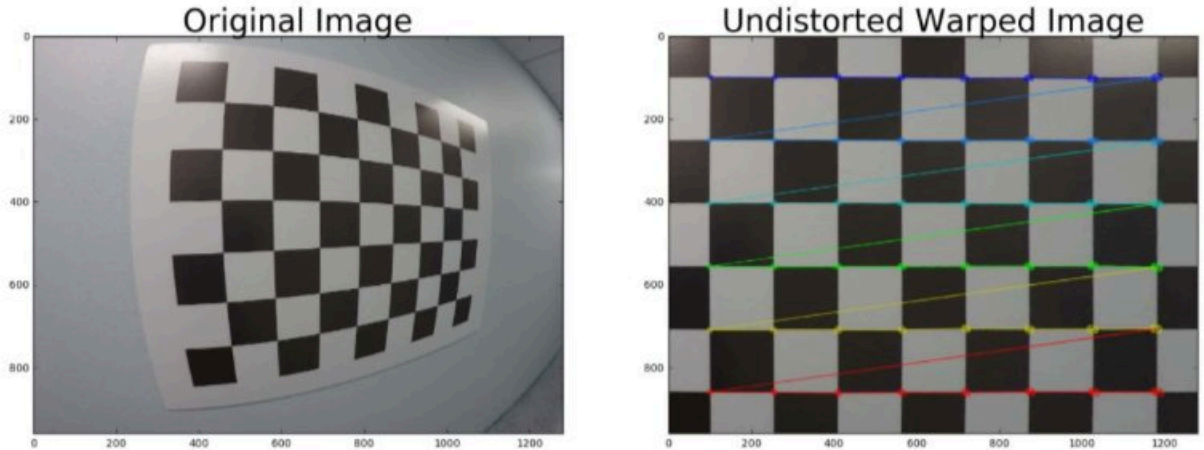




Top: `src = [[577,463],[707,463],[1110,720],[205,720]]` → `dst = [[300,0],[980,0],[980,720],[300,720]]`. Bottom: lane pixels in the bird's-eye view, second-order fits and the radius in **metres** (assuming 30 m per 720 px vertically and 3.7 m per 700 px horizontally; Lab 6D).

Quiz: undistort and transform a chessboard

▶ Lab 6D in Colab

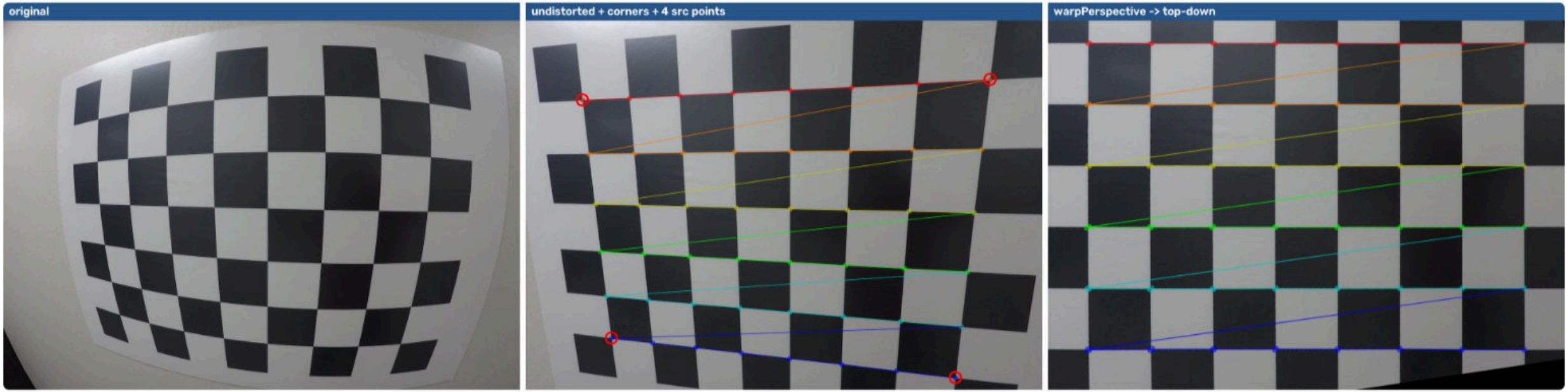


Given `mtx` and `dist`, write `corners_unwarp(img, nx, ny, mtx, dist)`:

1. Undistort with `cv2.undistort`
2. Convert to grayscale and find the chessboard corners; draw them
3. **src** = the **outer 4 detected corners**
4. **dst** = 4 points of a rectangle, **in the same order** as src
5. `getPerspectiveTransform` → M; `warpPerspective` → top-down view

```python
def corners_unwarp(img, nx, ny, mtx, dist):
    undist = cv2.undistort(img, mtx, dist, None, mtx)
    gray = cv2.cvtColor(undist, cv2.COLOR_BGR2GRAY)
    ret, corners = cv2.findChessboardCorners(gray, (nx, ny),
    M, warped = None, undist
    if ret:
        corners = corners.reshape(-1, 2)   # (N,1,2) in 4.x,
        cv2.drawChessboardCorners(undist, (nx, ny), corners,
        offset = 100
        w, h = gray.shape[1], gray.shape[0]
        src = np.float32([corners[0], corners[nx - 1],
                          corners[-1], corners[-nx]])
        dst = np.float32([[offset, offset], [w - offset, off
                          [w - offset, h - offset], [offset,
        M = cv2.getPerspectiveTransform(src, dst)
        warped = cv2.warpPerspective(undist, M, (w, h))
    return warped, M
```

Undistort & transform: the result



Our solution on a GoPro image from Lab 6. Note the order: **undistort first**. A homography maps straight lines to straight lines, so it cannot remove lens curvature.

OpenCV 5 note: `findChessboardCorners` returns corners of shape **(N, 2)**; OpenCV 4.x returned **(N, 1, 2)**. Code like `corners[0][0]` breaks. Use `corners.reshape(-1, 2)` to work with both.

Quiz: match each process to its purpose

MATCHING CHALLENGE

Processes: ① Camera calibration · ② Distortion correction · ③ Perspective transform

Purposes:

a. To transform an image so that we are effectively viewing objects from a different angle or direction.

b. To ensure that the geometrical shape of objects is represented consistently, no matter where they appear in an image.

c. To compute the transformation between 3-D object points in the world and 2-D image points.

d. To measure the orientation angles of objects in an image. (distractor)

① **Camera calibration → C**: computes the mapping between 3-D world points and 2-D image points.

② **Distortion correction → B**: shapes look the same anywhere in the frame.

③ **Perspective transform → A**: view the scene from a different angle, e.g. from above.

D is not a purpose of any of the three on its own.

REMEMBER

Perspective transform: the 3 things to remember

1. **Undistort first**, then warp: a homography keeps straight lines straight, so it cannot fix lens curves.
2. 4 point pairs → `getPerspectiveTransform` → `warpPerspective`.
3. Bird's-eye view → fit $x = Ay^2 + By + C$ → lane curvature in metres.

Watch & learn: recommended videos



Intrinsics & extrinsics
Cyrill Stachniss · 5 min



Intrinsic camera calibration
Cyrill Stachniss · 5 min



Zhang's method
Cyrill Stachniss · lecture



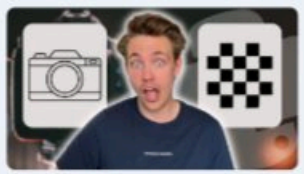
Camera calibration explained, step by step
Kevin Wood



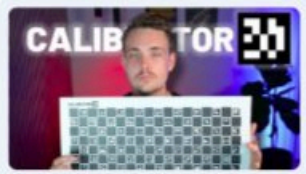
OpenCV Python calibration (intrinsic, extrinsic, distortion)
Kevin Wood



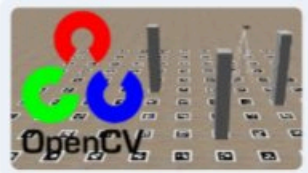
Camera calibration in < 5 minutes
Nicolai Nielsen



Calibration with a Python script
Nicolai Nielsen



ChArUco board calibration
Nicolai Nielsen



Calibrate an OpenCV camera in 15 minutes
The Drone Dojo (Raspberry Pi)

📖 Step-by-step tutorials for all 9 videos | Video labs: 6I Pi in 15 min | 6J Nicolai | 6K Kevin Wood | 6L ChArUco | 6M Zhang (Cyrill)

HANDS-ON · VIDEO LAB 6I

Calibrate your Raspberry Pi camera in 15 minutes

```
# on the Pi (SSH + VNC), Raspberry Pi OS Bookworm
mkdir -p ~/calib && cd ~/calib
for f in pi_calibrate_20.py aruco_pose_test.py; do
  wget https://raw.githubusercontent.com/Kazimbalti/\
intelligent-vision-systems/main/LAB/Lab_6/$f; done
wget https://raw.githubusercontent.com/opencv/opencv/4.x/doc/pattern.png
sudo apt install -y python3-opencv python3-picamera2

# 1. calibrate: 26 mm squares, 640 x 480 (as in the video)
python3 pi_calibrate_20.py --mm 26 --width 640 --height 480

# 2. print a marker, measure it, test with a ruler
python3 aruco_pose_test.py --make-marker --id 72
python3 aruco_pose_test.py --size 200 --id 72 \
        --width 640 --height 480
```

1. Print **pattern.png** on A4, glue it flat, **measure** a square
2. The script keeps **20 good, different** frames by itself: show the **edges** first, then near, far, tilted
3. It saves `cameraMatrix.txt`, `cameraDistortion.txt`, `camera.yaml`
4. **Ruler test:** centre an ArUco marker (x ≈ 0), slide it **6 cm**: the program must say **≈ 60 mm**
5. Same resolution in every script!

After The Drone Dojo: Calibrate OpenCV Camera In Under 15 Minutes, updated for Bookworm + OpenCV 5 (Picamera2, apt instead of sudo pip, solvePnP instead of estimatePoseSingleMarkers).

▶ Lab 6I in Colab | **Step-by-step tutorial**

Questions students always ask

Do I calibrate every time I switch the camera on?
No. Once per camera + lens + focus + resolution. Save `camera.yaml` and reuse it.

I changed the resolution. Recalibrate?
Best yes. Same aspect ratio: just scale f_x, f_y, c_x, c_y by the resize factor (dist stays the same).

Two cameras of the same model: same K?
Close, but not equal. Every unit is assembled slightly differently, so calibrate each one.

Why a chessboard and not any picture?
We must know the true shape exactly, and chessboard corners can be found to 0.1 px.

How many photos?
3 is the mathematical minimum; use **15–30** sharp, tilted views covering the whole frame.

My RMS is 0.3 px. Am I done?
Almost: also check the board covered the edges, and do a ruler test (Lab 6I): does a known distance come out right?

Summary

Model

- $s \cdot x = K\,[R \mid t]\,X$: **K** ($f_x$, f_y, c_x, c_y) is the camera, **[R | t]** is where it is
- Lenses add **radial** (k1, k2, k3) and **tangential** (p1, p2) distortion, applied in normalised coordinates
- Wide and fisheye lenses need k3 / rational or the **fisheye** model; stereo needs **R, T** between cameras

Practice

- `findChessboardCorners` (+ `cornerSubPix`) → `calibrateCamera` → RMS < 0.5–1 px
- 20+ sharp, tilted views covering the **whole frame**; same resolution & focus as deployment
- `undistort` or `initUndistortRectifyMap` + `remap` for video
- **Undistort first**, then `getPerspectiveTransform` + `warpPerspective` → bird's-eye → curvature

Next: Lecture 10 builds on this with epipolar geometry, disparity and depth from stereo, and the Lab 9 measuring system needs your calibrated camera.

Lab 6: all notebooks (click to open in Colab)

6A	Pinhole projection & distortion	6B	Finding chessboard corners
6C	Calibrate & undistort (GoPro)	6D	Road camera: bird's-eye & curvature
6E	ChArUco & fisheye (known truth)	6F	Bug hunt 🐞 (pairs)
6G	Stereo calibration & rectification	6H	Calibrate your own Pi / USB camera
6I	🎬 Pi camera in 15 minutes	6J	🎬 Get images → calibrate (Nicolai)
6K	🎬 Calibration explained (Kevin Wood)	6L	🎬 ChArUco pipeline (Nicolai)
6M	🎬 Zhang from scratch (Cyrill)	📋	Labs page: Colab · View · Download for every notebook

Scripts: pi_calibrate_20.py · aruco_pose_test.py · capture_calib.py · calibrate.py · get_images.py · all on GitHub · video tutorials

Reading: OpenCV calib3d · ArUco/ChArUco calibration · ROS camera_calibration · Zhang, TPAMI 2000 · Szeliski 2e §2.1, §11.1 · Medium: checkerboard calibration · udacity/CarND-Camera-Calibration

Credits. Adapted and extended from the Camera Calibration lesson of the Udacity Self-Driving Car Engineer Nanodegree. Lesson illustrations and calibration images © Udacity, Inc. (MIT-licensed repositories); stereo sample images from OpenCV (Apache 2.0). Videos © their authors. Used here for teaching.

Questions?

Next: Lab 6. Calibrate the GoPro, the road camera, a stereo pair, and then your own camera.